Carbon Taxes and the Inflation–Output Trade-off in the Low-Carbon Transition: How Financial Frictions Shape the Monetary Policy Response*

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Abstract

How should central banks respond when carbon taxes reshape the economy? This paper studies monetary policy during the low-carbon transition in a two-sector model in which carbon taxation strands carbon-intensive assets, weakens bank balance sheets, and amplifies the downturn through tighter credit—making the transition a financial shock as well as a supply shock. Comparing interest-rate rules, I find that aggressively easing policy as emissions fall can raise household welfare by fueling an investment boom, but only at the cost of inflation volatility incompatible with price-stability mandates. A modest response to emissions, by contrast, improves stabilization while leaving the emissions path—determined by fiscal policy—essentially unchanged. Emissions data are best used as a timely indicator of transition-related supply pressures, not as a policy target. Central banks can smooth the financial strains of decarbonization, but steering the transition itself should remain a fiscal task.

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1 Introduction

Climate change is an urgent global challenge that requires immediate action from policymakers, economists, and stakeholders. Within macroeconomics and finance, transition risks have gained prominence as economies shift toward a low-carbon future. These risks stem from the structural transformations from climate policies, technological advances, and evolving consumer preferences (Feyen et al., 2020). As governments implement increasingly ambitious climate policies to meet Paris Agreement commitments, central banks confront a fundamental question: how should monetary policy respond to carbon pricing shocks, and does the answer depend on how these policies are implemented?

The transition to a low-carbon economy presents significant risks of economic and financial instability, particularly for carbon-intensive sectors that rely heavily on fossil fuels. Industries such as oil, gas, coal, steel, aluminum, cement, and greenhouse horticulture are especially vulnerable to asset devaluations, potentially leading to stranded assets and severe financial repercussions (Van der Ploeg and Rezai, 2020). These disruptions could trigger rising debt defaults, speculative bubbles in emerging industries, and abrupt asset revaluations, creating systemic risks that extend across the broader economy (Semieniuk et al., 2021). Financial feedback effects can further amplify macroeconomic impacts, with expectations playing a critical role. As capital is reallocated, carbon-intensive firms may experience declining profitability, asset devaluation, and tighter credit conditions, while low-carbon firms benefit from improved financing (NGFS, 2024).

Climate policy and monetary policy are interconnected. Carbon taxes operate as supply shocks that raise production costs in emission-intensive sectors, creating inflationary pressures while simultaneously depressing output, a classic policy trade-off for central banks—one whose severity depends on how the policy is designed and how its revenues are recycled (Blanz et al., 2025). Schnabel (2022) has acknowledged that climate change could force it to take 'Greenflation' into account in monetary policy operations, while Diluiso et al. (2021) emphasize that mitigation policies, like carbon pricing, can be a direct source of shocks, creating potential trade-offs for monetary policy. Yet despite this growing policy attention, the academic literature provides limited guidance on how central banks should optimally calibrate their interest rate response to carbon tax transitions, and whether this response should depend on how climate policies are communicated and implemented.

According to the Intergovernmental Panel on Climate Change (Lee et al., 2023), there is a significant gap between the emissions reductions achieved under current policies and those required to meet the Paris Agreement targets, prompting the need for additional policy action by governments. From a macroeconomic perspective, beyond the direct impacts of climate mitigation policies, this policy gap introduces uncertainty about the transition pathway, which can affect investment decisions and long-term economic planning. To effectively manage these transition risks, it is essential to integrate them into macroeconomic models that account for financial frictions and environmental externalities.

In this context, I employ a two-sector Environmental Dynamic Stochastic General Equilibrium (E-DSGE) model with financial frictions, building on Gertler and Karadi (2011). The model incorporates Epstein-Zin preferences, and environmental externalities through a brown sector that emits pollutants and a green sector that operates sustainably. Firms in both sectors issue bonds to finance capital, with banks holding these bonds under leverage constraints, creating a spread between bond yields and deposit rates. The model includes two features that prove central to the results: endogenous abatement technology that allows brown firms to reduce emissions in response to carbon pricing, and financial frictions in the banking sector that can amplify macroeconomic fluctuations during the transition. This sectoral distinction is also consistent with recent empirical evidence that monetary policy transmission is not carbon-neutral: carbon-intensive firms display greater sensitivity to monetary policy shocks than greener firms, even after controlling for financial constraints (Benchora et al., 2025), so a monetary authority navigating the transition is de facto operating on a sectorally asymmetric economy of precisely the kind modeled here.

A distinctive feature of the framework is the adoption of recursive Epstein–Zin preferences in place of the expected-utility (CRRA) specification standard in most E-DSGE studies. The reason is economic rather than technical. The green transition confronts households with long-horizon risk: policy announcements, stranded-asset revaluations, and financial-sector stress affect not only current consumption but the entire distribution of future consumption paths. Under CRRA utility, a single parameter is forced to govern both the aversion to this risk and the willingness to shift consumption over time, mechanically tying the demand for precautionary savings to the intertemporal response to interest rates. Epstein–Zin preferences break this link: households can be highly averse to transition risk while remaining willing to substitute consumption intertemporally. This separation is what allows the analysis to distinguish the *risk channel* of monetary policy—smoothing uncertainty about the transition path—from the standard *intertemporal channel*, and, as shown below, it is the elasticity of intertemporal substitution rather than risk aversion that governs whether monetary accommodation stimulates investment or merely fuels inflation. The quantitative contribution of this preference structure is assessed directly: Section 5 re-solves the model under CRRA utility, and the robustness analysis maps the results over the full range of risk-aversion and substitution parameters, including configurations in which households prefer late resolution of uncertainty.

I characterize optimal policy by searching the parameter space of Taylor-type rules. The baseline specification is the standard Taylor rule with responses to inflation (ϕ_π) and output (ϕ_y). I then consider, normatively, an *emissions-augmented* Taylor rule that adds a response to emissions growth (ϕ_e), following proposals in the literature (Chen et al., 2021; Ramlall, 2023). For each rule specification and scenario, I identify parameters that maximize welfare, minimize the central bank's loss, or minimize cumulative emissions, characterizing the trade-offs policymakers face. The reference experiment is a *permanent*, unanticipated carbon-tax increase—reflecting the intended permanence of climate policy—around which I evaluate two alternative implementation paths: a persistent but imperfectly credible surprise, and a credibly announced gradual phase-in.

Relative to the existing literature, the contribution is threefold. First, whereas prior work on climate policy and financial frictions focuses on macroprudential instruments (Carattini et al., 2023) or characterizes the stagflationary trade-off under fixed policy rules (Del Negro et al., 2023; Nakov and Thomas, 2026), this paper *optimizes* simple, implementable Taylor rules—standard and emissions-augmented—across welfare, stabilization, and emissions objectives, under a permanent carbon-tax benchmark as well as alternative implementation paths. Second, the interaction between the financial accelerator and endogenous abatement is new: financial frictions determine how much of the carbon-tax shock is amplified through bank balance sheets, while the abatement margin determines how much of the adjustment occurs without contracting output; together they define the scope for monetary accommodation. Third, the recursive-preference structure delivers a sharp policy result: the welfare-maximizing rule is structurally incompatible with an inflation-targeting mandate, while a small emissions response—using emissions as an informative supply-side indicator—improves stabilization at negligible inflationary cost. Each ingredient is quantified separately: the paper reports results without financial frictions, without abatement, and under CRRA preferences, so the reader can attribute the findings to specific mechanisms rather than to the joint calibration.

The results reveal that the optimal standard Taylor rule takes a relatively weak stance on inflation, providing only partial accommodation to the transition's cost-push shock. Introducing an emissions response into the rule yields welfare gains of 0.02–0.04 percentage points in consumption-equivalent terms—largest under the permanent benchmark—though this benefit comes at the cost of substantially higher inflation volatility. From a stabilization perspective, however, the optimal strategy shifts; minimizing central bank loss calls for a small but positive emissions coefficient, suggesting that emissions data can serve as a valuable supply-side indicator without destabilizing prices. This policy ranking remains consistent across the permanent benchmark and both transition-path scenarios, with permanence raising the welfare stakes and anticipation magnifying both the welfare benefits and the inflationary costs of such accommodation.

This paper contributes to the debate on Central Banks' roles in climate risk management (Battiston et al., 2017; Campiglio et al., 2018), suggesting that the optimal monetary response to carbon pricing is neither to ignore the transition shock nor to actively steer it. Rather than adopting an explicit climate mandate — a step that risks overburdening monetary policy in violation of the Tinbergen principle and creating tensions with traditional central bank objectives (Chen et al., 2021; Masciandaro and Russo, 2024) — central banks can improve transition outcomes by treating emissions dynamics as an informative signal of underlying supply conditions. This "pragmatic augmentation" yields some welfare gains while preserving price stability, consistent with recent arguments that monetary policy should "see through" the transitory cost-push component of carbon pricing rather than either accommodating or fighting it aggressively (Olovsson and Vestin, 2026; Del Negro et al., 2023; Schnabel, 2022).

At the same time, the finding that aggressive emissions targeting generates large welfare improvements at the cost of severe nominal instability emphasizes the importance of institutional design: the welfare-optimal policy is structurally incompatible with inflation-targeting mandates, reinforcing the case for a clear division of roles in which fiscal instruments drive the emissions path, and monetary policy drives the macroeconomic adjustment (McKibbin et al., 2020; Nakov and Thomas, 2026; Coenen et al., 2024).

The remainder of the paper is organized as follows. Section 2 presents the related literature. Section 3 presents the quantitative model. Section 4 presents the Computational Strategy, and

Section 5 presents the main results, including optimal parameters, welfare comparisons, and impulse-response analysis. Section 6 delivers a discussion about the main results and policy implications. Section 7 concludes.

2 Related Literature

There is a growing body of macroeconomic research that employs Dynamic Stochastic General Equilibrium (DSGE) models with environmental externalities to examine the relationship between climate policies and business cycles. These models, commonly referred to as Environmental DSGE (E-DSGE) models,[1] provide a structured framework for assessing the macroeconomic implications of transitioning to a low-carbon economy. This paper aligns with those that incorporate financial frictions to analyze the effects of climate policies in the presence of credit market imperfections, with a particular focus on the role of conventional monetary policy.

A strand of this literature examines how financial frictions interact with climate policy. Carattini et al. (2023) and García-Villegas and Martorell (2024) show that carbon pricing can generate significant financial stability risks through the banking sector—the former in an E-DSGE model where even moderate carbon tax shocks trigger financial instability, the latter demonstrating that carbon taxation spills over into bank balance sheets and that capital requirements can contain these transition risks. In both cases, macroprudential instruments emerge as the tool to mitigate the financial fallout. My contribution is complementary but distinct in three respects. First, the focus is on *monetary* rather than macroprudential instruments. Second, I systematically optimize Taylor rule parameters across multiple objectives rather than analyzing fixed baseline rules. Third, the model incorporates endogenous abatement technology, which proves central to understanding the scope for monetary accommodation: when firms can reduce emissions intensity in response to carbon pricing, part of the adjustment occurs at the intensive margin, reducing the burden on aggregate demand management.

The financial-amplification mechanism at the core of this paper is not specific to the DSGE tradition. In stock-flow-consistent and agent-based frameworks, Dafermos et al. (2018) show that climate-related damages and asset repricing can erode corporate liquidity and bank capital, gener-

[1] For an extensive review of E-DSGE models and business cycles, see Annicchiarico et al. (2022).

ating credit contractions that feed back into real activity, and argue that monetary and financial policies shape the severity of this loop. More recently, Fierro et al. (2026) find, in a large-scale integrated macro-financial model, that rapid carbon pricing and energy transitions can threaten macro-financial stability unless accompanied by supportive policy, while orderly transitions keep financial stress contained. The present results are consistent with these findings from a complementary, optimizing-model perspective: the transition operates through bank balance sheets, and the stance of monetary policy determines whether the financial system amplifies or absorbs the shock. The value added of the DSGE implementation is the ability to conduct welfare analysis and to optimize the policy rule explicitly, at the cost of a more stylized financial sector.

On conventional monetary policy and climate, McKibbin et al. (2020) analyze how different monetary frameworks respond to climate change, finding that inflation targeting exacerbates output declines under carbon taxes while nominal income targeting mitigates these effects. Their results stress the importance of coordinating climate and monetary policy, as improper monetary responses to carbon pricing can heighten macroeconomic volatility. Chen et al. (2021) introduce a climate-augmented monetary policy rule within an E-DSGE model, finding that monetary policy must account for existing climate policies. While they explore emissions-gap targeting, which can enhance welfare but risks overburdening central banks, this paper takes a different approach by systematically comparing Taylor rule specifications across different carbon tax implementation scenarios and identifying the structural conditions under which emissions targeting is beneficial.

Recent literature has characterized the green transition primarily through the lens of production networks and relative price adjustments. Del Negro et al. (2023) employ a granular multi-sector framework to demonstrate that carbon taxation generates a sizable, albeit transitory, stagflationary trade-off. Regarding the policy response, Olovsson and Vestin (2026) argue analytically that central banks should "look through" these energy price changes by targeting core inflation, while Nakov and Thomas (2026) find that under realistic (suboptimal) carbon tax paths, monetary policy remains a tool that should prioritize strict price stability. I complement this literature by introducing a critical transmission channel absent in these frameworks: the financial sector. By modeling the feedback loop between asset prices, bank net worth, and investment, I show that the transition is not merely a supply shock but also a financial shock. Consequently, optimal policy depends not just on the level of the carbon tax, but also on its timing (anticipated vs. surprise) and on the degree to which monetary accommodation can shield the financial sector from asset stranding while incentivizing endogenous abatement.

My focus on anticipated versus unanticipated policy implementation connects the news shocks literature (Beaudry and Portier, 2006; Schmitt-Grohé and Uribe, 2012) to recent applications in climate economics. For instance, Bartocci et al. (2024) show that the inflationary impact of carbon taxes depends critically on the timing of household expectations, finding that anticipation can amplify near-term inflationary pressures. Similarly, Coenen et al. (2024) show that the macroeconomic costs of the transition are highly sensitive to the credibility of the policy and the monetary reaction function. I complement these studies by optimizing policy rules across distinct implementation scenarios—rather than analyzing fixed baselines—and by explicitly identifying the role of financial frictions and endogenous abatement in propagating these 'climate news' shocks.

This paper also contributes to the literature on whether central banks should adopt expanded mandates. While the traditional view asserts that central banks should focus solely on price stability (Clarida et al., 1999; Svensson, 2000), Debortoli et al. (2019) argue that loss functions should place significant weight on economic activity. This debate has gained renewed relevance as central banks face transition risks (Acharya, 2015; Krogstrup and Oman, 2019). I show that the optimal policy specification depends on how climate policy is implemented and on the economy's structural features, suggesting that mandate considerations should be context-dependent rather than universal.

3 Model

I develop a two-sector Environmental Dynamic Stochastic General Equilibrium (E-DSGE) model with financial frictions to analyze optimal monetary policy during carbon tax transitions. The model builds on elements from Diluiso et al. (2021) and Ferrari and Nispi Landi (2024), featuring a New Keynesian framework with financial intermediation, a representative household, and two production sectors: a green sector that operates sustainably and a brown sector that emits pollutants. The economy incorporates sector-specific capital inputs and multiple bank-held assets, with financial frictions arising from an agency problem that limits banks' ability to obtain funds from households, following Gertler and Karadi (2011). Nominal price rigidities make monetary policy central to real activity dynamics and the transmission of shocks. The sectoral structure enables the analysis of systemic spillover effects from climate policy and the associated transition risks.

Additionally, I adopt recursive Epstein–Zin preferences (Epstein and Zin, 1991; Barrage, 2020), which relax the tight link between risk aversion and the elasticity of intertemporal substitution implied by standard CRRA utility. This flexibility is particularly important in the context of climate policy, where households face long-horizon uncertainty about both economic outcomes and environmental damages. The recursive structure allows agents to form forward-looking assessments of welfare risk, amplifying the welfare consequences of carbon taxation and monetary policy responses to transition shocks. A growing literature in climate economics motivates this specification, showing that Epstein–Zin preferences can generate optimal policies and welfare implications that differ sharply from those obtained under expected utility and the Ramsey framework (Ackerman et al., 2013; Hambel et al., 2021).

Then, the model features endogenous abatement technology that allows brown firms to reduce emissions in response to carbon pricing. This margin of adjustment proves important for understanding optimal monetary policy. When firms can abate, they absorb part of the carbon tax shock by investing in cleaner production, dampening the emissions response, and reducing the burden on monetary policy. Additionally, financial frictions in the banking sector can amplify macroeconomic fluctuations during the transition.

3.1 Households

There is a continuum of infinitely lived households with preferences over consumption and leisure. Each household consists of workers and bankers. Workers supply labor and return their wages to the household, while bankers manage financial intermediaries and transfer profits to the household. Households cannot lend to intermediaries owned by their own members. As in Gertler and Karadi (2011), a fraction $1 - f$ of household members are workers and the remaining fraction f are bankers, with individuals transitioning between occupations over time at an exogenous rate that keeps proportions constant.

The representative household has recursive Epstein-Zin preferences over consumption (c_t) and leisure ($1 - L_t$):

$$V_t = \left[(1-\beta)\left(\left((c_t - h\, c_{t-1})^{\omega}\,(1-L_t)^{1-\omega}\right)^{1-\rho}\right) + \beta\left[\mathbb{E}_t\left(V_{t+1}^{1-\sigma}\right)\right]^{\frac{1-\rho}{1-\sigma}}\right]^{\frac{1}{1-\rho}}. \tag{1}$$

where ω is the utility weight on consumption, β is the discount factor, $\sigma > 0$ is risk aversion, and $\rho = 1/\psi$ with $\psi > 0$ being the elasticity of intertemporal substitution. The labor composite is $L_t = \left(L_{b,t}^{1+\rho_l} + L_{g,t}^{1+\rho_l}\right)^{\frac{1}{1+\rho_l}}$, where ρ_l governs the elasticity of substitution between sectors.

The household budget constraint is:

$$c_t + D_{t+1} = w_{b,t}L_{b,t} + w_{g,t}L_{g,t} + T_t + R_tD_t + \Pi_{g,t} + \Pi_{b,t} + \Phi_t \tag{2}$$

where $w_{k,t}$ is the real wage in sector k, D_t are deposits earning gross return R_t, T_t are lump-sum taxes/transfers, $\Pi_{k,t}$ are profits from sector k firms, and Φ_t are transfers from exiting bankers that serve as start-up funds for new bankers.

The first-order conditions yield labor supply equations for each sector:

$$(1-L_t)\left(L_{b,t}^{1+\rho_l} + L_{g,t}^{1+\rho_l}\right)^{\frac{\rho_l}{1+\rho_l}} L_{k,t}^{-\rho_l} = \frac{1-\omega}{\omega}\frac{c_t - hc_{t-1}}{w_{k,t}}, \quad k \in \{g, b\}. \tag{3}$$

The household's stochastic discount factor $M_{t,t+1}$ inherits the recursive structure of preferences: in addition to the standard consumption-growth component, it contains a forward-looking adjustment term in the continuation value V_{t+1}, which raises the price of assets that pay off badly when long-run prospects deteriorate. This is the channel through which transition risk is priced. The full expression is derived in Appendix B.4; the Euler equation is $\mathbb{E}_t\left[M_{t,t+1}R_{t+1}\right] = 1$.

3.2 Final Goods Producers

A representative final goods firm aggregates intermediate goods using a CES technology:

$$Y_t = \left[\int_0^1 y_t(i)^{\frac{\epsilon-1}{\epsilon}}\, di\right]^{\frac{\epsilon}{\epsilon-1}} \tag{4}$$

where $y_t(i)$ is the intermediate good produced by firm i and ϵ is the elasticity of substitution.

Profit maximization yields the demand function:

$$y_t(i) = Y_t \left(\frac{p_t(i)}{P_t} \right)^{-\epsilon} \tag{5}$$

where $p_t(i)$ is the price set by firm i and P_t is the aggregate price level.

3.3 Intermediate Producers

A continuum of intermediate firms produce differentiated goods using a CES bundle of green ($y_{g,t}$) and brown ($y_{b,t}$) inputs:

$$y_t(i) = \left[(1-\zeta)^{\frac{1}{\gamma}} y_{g,t}(i)^{\frac{\gamma-1}{\gamma}} + \zeta^{\frac{1}{\gamma}} y_{b,t}(i)^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}} \tag{6}$$

where ζ is the share of brown goods in production and γ is the elasticity of substitution between green and brown inputs. Cost minimization yields demand functions:

$$y_{g,t} = (1-\zeta) \left(\frac{p_{g,t}}{MC_t} \right)^{-\gamma} y_t, \quad y_{b,t} = \zeta \left(\frac{p_{b,t}}{MC_t} \right)^{-\gamma} y_t \tag{7}$$

where $MC_t = \left[(1-\zeta) p_{g,t}^{1-\gamma} + \zeta p_{b,t}^{1-\gamma} \right]^{\frac{1}{1-\gamma}}$ is the marginal cost.

In this monopolistic competition setting, firms face quadratic price adjustment costs à la Rotemberg (1982). From a symmetric equilibrium, I obtain the New Keynesian Phillips Curve:

$$\pi_t(\pi_t - \bar{\pi}) = \mathbb{E}_t \left[M_{t,t+1} \pi_{t+1} (\pi_{t+1} - \bar{\pi}) \frac{Y_{t+1}}{Y_t} \right] + \frac{\epsilon}{\kappa_P} \left(MC_t - \frac{\epsilon - 1}{\epsilon} \right) \tag{8}$$

where π_t is inflation, $\bar{\pi}$ is steady-state inflation, and κ_P is the price adjustment cost parameter.

3.4 Green and Brown Producers

Both sectors produce using Cobb-Douglas technology with capital and labor:

$$y_{k,t} = A_t (\xi_{k,t} K_{k,t-1})^{\alpha} L_{k,t}^{1-\alpha}, \quad k \in \{g, b\} \tag{9}$$

where A_t is aggregate productivity affected by pollution through a damage function standard in DICE models (Nordhaus, 2017): $A_t = (1 - (D_0 + D_1 X_t + D_2 X_t^2))a_t$, with a_t following an AR(1) process. The term $\xi_{k,t}$ is a sector-specific capital quality shock.

Following Heutel (2012), only the brown sector generates emissions. The stock of pollution evolves according to:

$$X_t = (1 - \delta_x)X_{t-1} + e_t + e^* \tag{10}$$

where e_t are emissions from brown production, δ_x is the natural decay rate, and e^* represents emissions from the rest of the world.

Emissions depend on brown output and the level of abatement:

$$e_t = (1 - ab_t)y_{b,t}^{1-\lambda} \tag{11}$$

where $ab_t \in [0, 1]$ is the abatement rate and λ governs the elasticity of emissions with respect to output.

Brown firms maximize profits by choosing labor, capital, and abatement. They finance capital purchases by issuing bonds $b_{b,t} = q_{b,t}K_{b,t}$ and face a carbon tax τ_t on emissions. Abatement is costly:

$$Z_t = \theta_1 ab_t^{\theta_2} y_{b,t} \tag{12}$$

where θ_1 and θ_2 govern abatement cost efficiency.

The brown firm's problem is:

$$\max \quad p_{b,t}y_{b,t} - w_{b,t}L_{b,t} - r_{kb,t}K_{b,t-1} - \tau_t e_t - Z_t$$

where $r_{kb,t} = r_{b,t}q_{b,t-1} - (1 - \delta)q_{b,t}$ is the rental rate of capital. The first-order conditions determine labor demand, capital demand, and optimal abatement:

$$(1 - \alpha)y_{b,t}\left(p_{b,t} - \tau_t(1 - ab_t)(1 - \lambda)y_{b,t}^{-\lambda} - \theta_1 ab_t^{\theta_2}\right) = w_{b,t}L_{b,t} \tag{13}$$

$$\alpha y_{b,t}\left(p_{b,t} - \tau_t(1 - ab_t)(1 - \lambda)y_{b,t}^{-\lambda} - \theta_1 ab_t^{\theta_2}\right) = r_{kb,t}K_{b,t-1} \tag{14}$$

The optimal abatement condition equates the marginal cost of abatement to the marginal benefit from avoided carbon taxes:

$$\theta_1\theta_2 a b_t^{\theta_2-1} = \tau_t y_{b,t}^{-\lambda} \tag{15}$$

Green firms face the same production structure but are not subject to carbon taxes or abatement costs:

$$(1-\alpha)y_{g,t}p_{g,t} = w_{g,t}L_{g,t}, \quad \alpha y_{g,t}p_{g,t} = r_{kg,t}K_{g,t-1} \tag{16}$$

3.5 Capital Producers

Capital producers purchase final goods and undepreciated capital to produce new capital for both sectors. They face investment adjustment costs and solve:

$$\max \quad \mathbb{E}_0 \sum_{t=0}^{\infty} M_{0,t} \sum_{k\in\{g,b\}} \left[q_{k,t}K_{k,t} - (1-\delta)q_{k,t}K_{k,t-1} - I_{k,t}\right]$$

$$\text{s.t.} \quad K_{k,t} = (1-\delta)K_{k,t-1} + \left[1 - \frac{\kappa_I}{2}\left(\frac{I_{k,t}}{I_{k,t-1}} - 1\right)^2\right] I_{k,t}$$

where $q_{k,t}$ is the price of capital in sector k, δ is depreciation, and κ_I governs adjustment costs. The resulting first-order condition—a standard Tobin's-q relation linking the price of capital to current and expected investment growth—is reported in Appendix B.4.

3.6 Financial Intermediaries

Financial intermediaries introduce financial frictions that can amplify macroeconomic fluctuations during the carbon transition. I follow Gertler and Karadi (2011), with the adaptations from Ferrari and Nispi Landi (2024) and Carattini et al. (2023) in modeling this sector.

Including financial frictions is crucial for understanding transition risk. When carbon-intensive sectors face policy shocks, the value of brown assets declines, reducing bank collateral. This deterioration in balance sheets leads to tighter credit conditions, higher financing costs, and potentially a contraction in economic activity. Carattini et al. (2023) show that an economy with financial frictions experiences a deeper recession than a frictionless economy, amplifying transition risks.

Banks raise deposits from households and lend to green and brown producers. The balance sheet of the representative bank is:

$$b^f_{g,t} + b^f_{b,t} = NW_t + D_t \tag{17}$$

where $b^f_{k,t}$ are bonds held on firms in sector k, NW_t is bank net worth, and D_t are deposits.

To generate a spread between green and brown bond rates, I assume banks face a quadratic cost when deviating from a target portfolio share b^*:

$$NW_t = r_{g,t}b^f_{g,t-1} + r_{b,t}b^f_{b,t-1} - \frac{r_{t-1}}{\pi_t}D_{t-1} - \frac{\kappa_F}{2}NW_{t-1}\left[\frac{b^f_{g,t-1}}{b^f_{t-1}} - b^*\right]^2 \tag{18}$$

This cost captures real-world frictions in portfolio reallocation, such as regulatory compliance costs or internal risk management policies. It creates a wedge between green and brown returns, preventing instantaneous adjustment to market conditions.

An agency problem limits bank leverage. Bankers can divert a fraction θ of assets, so depositors require that the bank's continuation value exceeds the diversion benefit:

$$V^f_t \geq \theta\left(b^f_{g,t} + b^f_{b,t}\right) \tag{19}$$

where the bank's value is:

$$V^f_t = \mathbb{E}_t\left[\sum_{i=0}^{\infty}(1-\chi)\chi^i M_{t,t+i}NW_{t+1+i}\right] \tag{20}$$

and χ is the probability a banker continues operating.

Assuming the constraint binds, the first-order conditions determine bank leverage $lev_t = b^f_t/NW_t$ and the portfolio split between green and brown bonds ($lev^g_t = b^f_{g,t}/NW_t$). Intuitively, leverage is increasing in the expected excess return on bank assets and decreasing in the diversion parameter θ, while the portfolio condition equates the expected green–brown return differential to the marginal portfolio adjustment cost. The full derivation, following the guess-and-verify approach of Ferrari and Nispi Landi (2024), together with the resulting leverage and portfolio conditions and the recursion for the bank's marginal value of net worth ν_t, is presented in Appendix B.3.

Aggregate bank net worth combines continuing and new bankers:

$$NW_t = \chi\left(\frac{r_{t-1}}{\pi_t} + \left(r_{b,t} - \frac{r_{t-1}}{\pi_t}\right) lev_{t-1} + (r_{g,t} - r_{b,t}) lev^g_{t-1} - \frac{\kappa_F}{2}\left(\frac{lev_{g,t-1}}{lev_{t-1}} - b^*\right)^2\right) NW_{t-1} + \iota b^f_t \tag{21}$$

where ι is the fraction of assets transferred to new bankers.

3.7 Monetary Policy

Monetary policy is conducted through Taylor-type interest rate rules with interest rate smoothing. In the baseline specification, the central bank follows a standard rule,

$$\frac{r_t}{\bar{r}} = \left(\frac{r_{t-1}}{\bar{r}}\right)^{\rho_r}\left[\left(\frac{\pi_t}{\bar{\pi}}\right)^{\phi_\pi}\left(\frac{Y_t}{\bar{Y}}\right)^{\phi_y}\right]^{1-\rho_r} e^{\varepsilon^{mp}_t}, \tag{22}$$

where r_t denotes the nominal policy rate, $\bar{r}$ its steady-state value, π_t inflation, $\bar{\pi}$ the inflation target, ρ_r captures interest rate smoothing, and $\varepsilon^{mp}_t \sim$ i.i.d. $N(0, \sigma^2_{mp})$ is a monetary policy shock. The coefficients ϕ_π and ϕ_y govern the policy response to inflation and output fluctuations induced by the carbon tax.

To study monetary policy responses to transition risk, I augment the rule to allow for systematic reactions to emissions dynamics:

$$\frac{r_t}{\bar{r}} = \left(\frac{r_{t-1}}{\bar{r}}\right)^{\rho_r}\left[\left(\frac{\pi_t}{\bar{\pi}}\right)^{\phi_\pi}\left(\frac{Y_t}{\bar{Y}}\right)^{\phi_y}\left(\frac{\sum_{j=1}^{4} e_{t-j}}{\sum_{j=5}^{8} e_{t-j}}\right)^{\phi_e}\right]^{1-\rho_r} e^{\varepsilon^{mp}_t}. \tag{23}$$

The coefficient ϕ_e measures the sensitivity of the policy rate to emissions momentum. A positive value implies policy accommodation when emissions decline, a salient feature of a low-carbon transition. Through this channel, monetary policy supports asset prices and mitigates balance-sheet stress as carbon-intensive activity contracts.

The rule targets the rate of change in emissions rather than deviations from a fixed steady state. During the transition, the economy converges toward a new, lower-emissions equilibrium, making the initial steady state an inappropriate benchmark. Focusing on emissions growth allows the central bank to manage the pace of adjustment without resisting the underlying structural shift. The backward-looking four-quarter windows imply that policy responds to a smoothed, lagged measure of emissions momentum rather than to contemporaneous emissions; note that the most recent observation entering the rule is dated $t-1$, so the specification does not impose a full publication delay (Section 5.11 discusses the implications of realistic reporting lags and measurement error for the value of the emissions signal).

Emissions are not treated as an independent policy objective, nor as a substitute for climate policy. Instead, they serve as a transition-specific indicator of supply-side pressure that is imperfectly captured by inflation or the output gap. Carbon pricing alters relative production costs across sectors, and these reallocations appear in emissions data before they are fully reflected in aggregate prices. A positive response to declining emissions therefore represents state-contingent accommodation of transition-related supply disturbances rather than an attempt to directly steer climate outcomes.

Finally, while Ramsey policy provides a theoretical welfare benchmark, it relies on full commitment to a state-contingent policy path, which is implausible in the presence of policy uncertainty and implementation lags associated with climate policy announcements. By optimizing simple and implementable Taylor rules, the analysis identifies robust monetary responses that perform well even when expectations cannot be perfectly managed during the pre-announcement phase of the transition.

3.8 Government and Market Clearing

The government implements the carbon tax τ_t and rebates revenues lump-sum to households through T_t.[2] The carbon tax follows one of the three transition paths described in Section 3.9: a permanent unanticipated increase (the reference scenario), a persistent surprise, or an anticipated gradual phase-in.

The final goods market clears:

$$Y_t = c_t + I_t + \frac{\kappa_F}{2} NW_{t-1} \left[\frac{b^f_{g,t-1}}{b^f_{t-1}} - b^* \right]^2 + \frac{\kappa_P}{2}(\pi_t - \bar{\pi})^2 Y_t + Z_t \tag{24}$$

[2] The assumption of lump-sum rebates provides a clean welfare benchmark by abstracting from distortionary tax interactions. Alternative recycling schemes—such as labor tax cuts or green investment subsidies—would alter both the fiscal multiplier of the carbon tax and potentially the optimal monetary response; Blanz et al. (2025) show that the performance of market-based climate policy depends crucially on the recycling scheme. Section 6 discusses the implications for monetary–fiscal coordination.

where $I_t = I_{g,t} + I_{b,t}$ is aggregate investment and Z_t is abatement costs.

Bond markets clear with $b_{k,t} = b^f_{k,t}$ for $k \in \{g, b\}$, and labor markets clear in each sector.

3.9 Carbon Tax Policy and Transition Scenarios

The driver of the low-carbon transition in the model is an exogenous carbon tax, τ_t, levied on the emissions of the brown sector. The tax raises brown firms' marginal costs and induces decarbonization through two margins: an extensive margin (a contraction of brown output) and an intensive margin (investment in abatement, which lowers emissions per unit of output). All revenues are rebated lump-sum to households, isolating the allocative effects of the policy from distributional ones.

The tax evolves according to

$$\tau_t = (1 - \rho_\tau)\bar{\tau} + \rho_\tau \tau_{t-1} + \varepsilon_t^\tau + \sum_{k=4}^{8} \psi_k \, \varepsilon_{t-k}^{\tau,news}, \tag{25}$$

where $\bar{\tau}$ is the initial carbon tax, ρ_τ governs persistence, ε_t^τ is an unanticipated innovation, and the news terms $\varepsilon_{t-k}^{\tau,news}$ capture a policy announced at date $t-k$ and phased in through the weights ψ_k ($\sum_k \psi_k = 1$), so that an announced reform is incorporated into the tax between quarters 4 and 8 after the announcement.

Permanence of the policy and the benchmark scenario. Climate policies are intended as permanent changes to the regulatory environment, not temporary interventions, and a mean-reverting tax could in principle distort the policy evaluation: if agents expect the tax to recede, forward-looking rules might exploit the anticipated reversal. To rule this out, the *reference scenario* of the analysis is a **permanent, unanticipated increase** in the carbon tax, implemented by setting $\rho_\tau = 0.9999$ and treating the reform as a single, one-off innovation (the recurring variance of the tax shock is switched off in this experiment, while productivity and monetary shocks are retained as background risk). At this value the tax is constant to numerical precision over any policy-relevant horizon—the half-life of the deviation exceeds 1,700 years—so the experiment is economically equivalent to a permanent doubling of the carbon price, while the stationarity of (25) preserves the validity of the second-order perturbation solution and the generalized impulse-response analysis used for welfare evaluation.[3]

Around this benchmark, I consider two alternative implementation scenarios. Unlike the permanent benchmark, both are mean-reverting by construction—they capture reforms that are, respectively, imperfectly credible and announced-but-not-locked-in, rather than alternative routes to the same long-run tax level:

1. **Persistent surprise ($\rho_\tau = 0.99$):** an immediate, unanticipated hike that decays very slowly (half-life of roughly 17 years). This scenario captures imperfect policy credibility: agents attach some probability to a long-run reversal of the policy, as observed in jurisdictions where carbon pricing has been repealed or diluted.

2. **Anticipated phase-in:** a credible announcement at $t = 0$, implemented progressively over quarters 4–8 through the news structure in (25), with weights $\psi_k = \{0.10, 0.20, 0.30, 0.25, 0.15\}$. This captures the implementation lags of climate legislation and creates a window for anticipatory reallocation of investment and consumption. The staggered structure also creates a "news trap": inflationary pressures from forward-looking pricing arise at $t = 0$, before the actual supply contraction at $t \geq 4$, testing the central bank's ability to look through the anticipation phase.

Comparing the three scenarios separates the role of *permanence* (benchmark vs. persistent surprise) from the role of *anticipation* (surprise vs. phase-in) in shaping the optimal monetary response. As shown in Section 5, the ranking of policy rules—and in particular the desirability of a small emissions response under a stabilization objective—is invariant across the three scenarios; what changes is the magnitude of the welfare stakes. An important reason for this invariance is that the augmented rule responds to emissions *growth* rather than to deviations from the initial steady state, so the policy prescription does not depend on whether the economy returns to the old equilibrium or converges to a new, lower-emissions one.

The shock size is chosen so that the carbon tax approximately doubles from its initial steady-state value, reaching approximately \$35 per ton of CO_2, consistent with the global average carbon

[3] The natural alternatives—a unit root, or a deterministic terminal steady state solved under perfect foresight—either invalidate the stochastic welfare analysis (which requires stationarity) or remove precisely the risk considerations that motivate the recursive-preference framework. The quasi-permanent specification retains both. A perfect-foresight transition, reported in Appendix C.3, yields real dynamics indistinguishable from the stochastic impulse responses over the first 40 quarters.

price in 2024 and within the range of carbon taxes implemented in Europe. The conversion follows Carattini et al. (2023), who map DSGE model units to dollar values using U.S. GDP and emissions data.[4] The horizon for welfare and emissions evaluation is set to 40 quarters, allowing sufficient time for the economy to adjust to the carbon tax level while remaining within a policy-relevant time frame.

3.10 Assessment Criteria

The central question of this paper is how the optimal values of ϕ_π, ϕ_y, and ϕ_e depend on: (i) how the carbon tax is implemented (permanent, persistent surprise, or anticipated phase-in); (ii) the availability of abatement technology; and (iii) the presence of financial frictions.

To evaluate monetary policy performance, I use different criteria. First, household welfare is measured by the recursive specification of Epstein-Zin preferences V_t, where I compute the change from the Stochastic Steady state in Consumption Equivalent Units.

Second, a central bank loss function following Debortoli et al. (2019):

$$L_t^{CB} = \lambda_\pi(\pi_t - \bar{\pi})^2 + \lambda_{\Delta r}(r_t - r_{t-1})^2 + \lambda_y(Y_t - \bar{Y})^2 \tag{26}$$

where the weights $\lambda_\pi = 1$, $\lambda_{\Delta r} = 0.077$, and $\lambda_y = 1.042$ reflect priorities consistent with dual-mandate central banks. The inclusion of interest rate volatility ensures gradual policy adjustments, while the weight on output acknowledges trade-offs in responding to supply shocks.

Comparing results under the welfare criterion and the central bank loss function reveals whether there are trade-offs between household welfare and traditional macroeconomic stabilization objectives. Such trade-offs are policy-relevant, as they indicate whether a central bank focused narrowly on its traditional mandate would choose different policies than one seeking to maximize household welfare.

Finally, as the goal of the transition is to move to a low-carbon economy, I assess the discounted sum of emissions over 10 years. With this complementary measure, one can assess whether there are trade-offs between welfare, macroeconomic stability, and emissions reduction

[4] While leading carbon pricing jurisdictions have exceeded €80/ton (EU ETS in 2023), the calibration targets the global average carbon price. The robustness analysis in Section 5 demonstrates that for larger shocks, the optimal emissions response converges toward zero as cost-push inflation dominates, suggesting aggressive accommodation is feasible only for moderate carbon price increases.

3.11 Calibration

The model is calibrated at a quarterly frequency for the U.S. economy. I draw on the extensive DSGE estimation literature, particularly Smets and Wouters (2007) and Gertler and Karadi (2011), supplemented with parameters from the environmental macroeconomics literature (Heutel, 2012; Carattini et al., 2023; Nordhaus, 2017).

Tables 1 and 2 summarize the complete calibration.

Table 1: Macroeconomic Parameters

Parameter	**Description**	**Value**	**Source**
Preferences			
β	Discount factor	0.995	Smets and Wouters (2007)
h	Consumption Habit	0.70	Smets and Wouters (2007)
σ	Risk aversion	10	Pohl et al. (2018)
ψ	Intertemporal elasticity of substitution	1.5	Pohl et al. (2018)
ω	Consumption utility weight	0.32	Barrage (2020)
Production			
α	Capital share	0.33	Smets and Wouters (2007)
δ	Capital depreciation rate	0.025	Smets and Wouters (2007)
ε	Elasticity of substitution	6	Christiano et al. (2005)
κ_p	Rotemberg price stickiness	58.53	Match calvo =0.75
κ_I	Investment adjustment cost	5.5	Smets and Wouters (2007)
Monetary Policy			
$\bar{\pi}$	Target inflation rate (quarterly)	1.005	2% Federal Reserve Target
ρ_r	Interest rate smoothing	0.80	Smets and Wouters (2007)
Financial Frictions			
χ	Banker survival probability	0.965	Gertler and Karadi (2011)
θ	Divertible asset share	Calibrated	Match $\phi_{ss} = 6$
ι	New banker wealth share (=0.003)	Calibrated	Match spread = 200 bps

Risk aversion is set to $\sigma = 10$ and $\psi = 1.5$ following the long-run risks literature (Pohl et al., 2018) and climate-economy models under uncertainty (Cai and Lontzek, 2019), implying precautionary savings and significant risk premia. The intertemporal elasticity of substitution ψ is set to 1.5 to capture empirically plausible consumption dynamics. Because $\sigma > 1/\psi$, this configuration implies a preference for early resolution of uncertainty, a feature of Epstein–Zin preferences that has attracted scrutiny in the literature on timing premia (Epstein et al., 2014): early-resolution agents demand more insurance against long-run risk, which could in principle predispose the model toward monetary accommodation. Section 5 therefore maps the optimal policy over the full (σ, ψ) space, including late-resolution configurations ($\sigma < 1/\psi$), and shows that the policy prescriptions do not depend on this feature of the calibration.

The elasticity of substitution across varieties $\varepsilon = 6$ implies a steady-state markup of 20%, following Christiano et al. (2005). Price stickiness is parameterized in Rotemberg form, with κ_p

set so that the slope of the Phillips curve matches a Calvo-equivalent price-change probability of 0.75—an average price duration of four quarters, consistent with Nakamura and Steinsson (2008).

The baseline Taylor rule parameters follow standard estimates: an interest rate smoothing $\rho_r = 0.80$ (Smets and Wouters, 2007), and the inflation target is the Federal Reserve's 2% annual rate.

Following Gertler and Karadi (2011), banker survival probability is $\chi = 0.965$, implying an average banker tenure of eight years. The divertibility parameter θ is calibrated to match a steady-state leverage ratio of 6, reflecting post-Dodd-Frank regulatory constraints. The wealth share for new bankers ι targets a corporate credit spread of 200 basis points annually.

The climate damage function follows the DICE model (Nordhaus, 2017):

$$D(X_t) = D_0 + \tilde{D}_1 X_t + \tilde{D}_2 X_t^2 \tag{27}$$

where X_t is the atmospheric CO_2 concentration in GtC. Following Ferrari and Nispi Landi (2024), I set $D_0 = -0.0076$, $\tilde{D}_1 = 8.10 \times 10^{-6}$, and $\tilde{D}_2 = 1.05 \times 10^{-8}$, which implies approximately 2.7% output loss at 3°C warming.

Table 2: Environmental and Sectoral Parameters

Parameter	**Description**	**Value**	**Source**
Climate Damages			
D_0	Damage constant	-0.0076	Nordhaus (2017)
$\tilde{D}_1$	Damage linear term	8.10×10^{-6}	Nordhaus (2017)
$\tilde{D}_2$	Damage quadratic term	1.05×10^{-8}	Nordhaus (2017)
δ_x	Pollution decay rate	0.0035	Nordhaus (2017)
Green-Brown Structure			
γ	Green-brown substitution	2	Carattini et al. (2023)
ζ	Brown goods weight	0.8	Giovanardi et al. (2023)
κ_G	Green capital share	0.20	U.S. renewable capacity
λ	Emissions-output elasticity	0.4	Heutel (2012)
e^{row}	Rest-of-world emissions ratio	8	Global emissions data
Abatement			
θ_1	Abatement cost scale	0.0334	Nordhaus (2017)
θ_2	Abatement cost curvature	2.6	Nordhaus (2017)

Green and brown goods exhibit imperfect substitutability with elasticity $\gamma = 2$ (Carattini et al., 2023). The initial green capital share is $\kappa_G = 0.20$, reflecting current renewable energy capacity in U.S. electricity generation. The emissions-output elasticity is $\lambda = 0.4$ (Heutel, 2012; Ferrari and Nispi Landi, 2024). The pollution decay rate $\delta_x = 0.0035$ quarterly corresponds to an atmospheric CO_2 half-life of approximately 50 years. The rest-of-world-to-U.S. emissions ratio is set to 8, so that

U.S. emissions account for one ninth ($\approx 11\%$) of the global pollution flow in steady state—consistent with total greenhouse-gas emissions of roughly 54 $GtCO_2e$ per year globally against about 6 $GtCO_2e$ for the United States.

3.12 Model Validation

Before turning to the policy analysis, I verify that the model generates plausible business-cycle dynamics. This subsection describes the procedure in full so the reader can assess exactly what is—and what is not—being matched.

Shock processes. The model features four disturbances: a TFP shock ($\rho_a = 0.95$, $\sigma_a = 0.01$; standard values from the estimated DSGE literature, Smets and Wouters, 2007), an i.i.d. monetary policy shock ($\sigma_{mp} = 0.0025$; magnitude from Smets and Wouters, 2007, with persistence in the policy rate arising from the interest-rate smoothing of the rule, $\rho_r = 0.8$, rather than from the shock process), and the carbon-tax innovations of equation (25). The validation simulations include only the two recurring business-cycle disturbances—TFP and monetary policy; the carbon-tax innovations are *excluded by design*, because they represent the policy experiments of Section 5 rather than a recurring feature of the business-cycle environment, and including them would conflate the two. None of the shock parameters is estimated to fit the moments reported below: persistence and volatility parameters are taken from the literature, and the model's fit should be read as a check of the propagation mechanism, not as the outcome of a moment-matching exercise. Table 3 summarizes the shock calibration and sources.

Table 3: Calibration of the Shock Processes

Shock	Process	Persistence	Std. dev.	Source
TFP (a_t)	AR(1)	0.95	0.010	Smets and Wouters (2007)
Monetary policy	i.i.d.	—	0.0025	Smets and Wouters (2007)
Carbon tax (surprise)	Eq. (25)	Sec. 3.9	0.010	Policy experiment
Carbon tax (news)	Eq. (25)	Sec. 3.9	0.010	Policy experiment

Note: The carbon-tax innovations enter only the transition experiments of Section 5; the validation simulations use the TFP and monetary-policy shocks.

Computation of model moments. The reported statistics are *simulated* (not theoretical population) moments, computed to mirror the treatment of the data. For each of 10,000 Monte Carlo replications, I simulate the model at the baseline calibration under a conventional Taylor rule ($\phi_\pi = 2$, $\phi_y = 0.08$, $\rho_r = 0.8$; Smets and Wouters, 2007) for 460 quarters, discard a 200-quarter burn-in, and retain 260 quarters—approximately the length of the empirical sample (251 quarters). Output, consumption, investment, and hours are logged and HP-filtered ($\lambda = 1600$); inflation and the nominal rate are demeaned. The "Model" column of Table 4 reports the mean statistic across replications, and the brackets report the 5th–95th percentile range of the *small-sample distribution* of each statistic. These bands therefore describe the sampling variability of a 260-quarter sample generated by the model; they are not confidence intervals for the data.

Empirical moments. The empirical counterparts use quarterly U.S. data over 1960Q2–2024Q4 (real GDP, consumption, and investment from the NIPA; nonfarm business hours; CPI inflation; and the federal funds rate), transformed identically (logs, HP(1600), demeaned rates). The sample excludes the COVID-19 years 2020–2021: those quarters are extreme outliers that mechanically inflate volatilities and distort autocorrelations, and their inclusion would degrade any model–data comparison for reasons unrelated to the propagation mechanisms studied here.[5]

Given the deliberately parsimonious two-shock environment, the model reproduces the qualitative hierarchy of volatilities (Panel A): investment is about three times as volatile as output, hours are somewhat more volatile than output (1.13 against 1.25 in the data), consumption is smoother (0.81 against 0.85), and the nominal variables are less volatile than output, with the relative volatility of inflation (0.43 against 0.58) captured reasonably well. Absolute volatilities are uniformly below the data (output standard deviation of 0.0112 against 0.0139), as expected when only two of the many disturbances driving observed fluctuations are included.

Regarding persistence and comovement (Panel B), the model performs well on the margins most relevant to the analysis: the autocorrelations of output (0.91 against 0.85) and consumption (0.88 against 0.85) are close to the data, with the model somewhat more persistent, and consumption and investment are strongly procyclical, as in the data. Investment persistence exceeds the data (0.94 against 0.82), reflecting the investment adjustment costs, and inflation is more persistent than in the data (0.70 against 0.50). The weakest dimensions are hours and the nominal comovements:

[5] The accompanying replication code reports the moments with the COVID years included; the largest effects are a sharp increase in the volatility of all real aggregates and a decline in their measured autocorrelations, driven by the 2020Q2 collapse and rebound.

Table 4: Business Cycle Moments: Empirical Data vs. Model

Panel A: Standard Deviations and Relative Volatilities

	Standard Deviation (σ)			**Relative SD** (σ_x/σ_y)		
Variable	Data	Model	[5%, 95%]	Data	Model	[5%, 95%]
Output (Y)	0.0139	0.0112	[0.0091, 0.0134]	1.00	1.00	—
Consumption (C)	0.0118	0.0090	[0.0075, 0.0106]	0.85	0.81	[0.73, 0.89]
Investment (I)	0.0625	0.0339	[0.0266, 0.0423]	4.48	3.03	[2.79, 3.29]
Hours (H)	0.0174	0.0124	[0.0111, 0.0138]	1.25	1.13	[0.95, 1.33]
Inflation (π)	0.0080	0.0047	[0.0040, 0.0057]	0.58	0.43	[0.33, 0.55]
Interest Rate (R)	0.0090	0.0045	[0.0035, 0.0059]	0.64	0.41	[0.31, 0.55]

Panel B: Persistence and Cyclicality

	Autocorrelation AR(1)			**Correlation with Output** Corr(x, Y)		
Variable	Data	Model	[5%, 95%]	Data	Model	[5%, 95%]
Output (Y)	0.85	0.91	[0.88, 0.93]	1.00	1.00	—
Consumption (C)	0.85	0.88	[0.85, 0.91]	0.86	0.93	[0.89, 0.95]
Investment (I)	0.82	0.94	[0.92, 0.95]	0.89	0.95	[0.93, 0.97]
Hours (H)	0.92	0.55	[0.46, 0.62]	0.84	0.20	[0.09, 0.30]
Inflation (π)	0.50	0.70	[0.59, 0.80]	0.14	-0.11	[-0.24, 0.02]
Interest Rate (R)	0.97	0.84	[0.75, 0.91]	0.18	-0.44	[-0.55, -0.32]

Note: Simulated moments: 10,000 Monte Carlo replications of 260 quarters (after a 200-quarter burn-in), HP(1600)-filtered logs for real variables and demeaned rates, mirroring the treatment of the data. The "Model" column reports the mean statistic across replications; the brackets [5%, 95%] report the small-sample distribution of each statistic across replications (not confidence intervals for the data).

hours are less persistent (0.55 against 0.92) and only mildly procyclical (0.20 against 0.84), and the policy rate is countercyclical in the model (-0.44 against $+0.18$). These misses share a common source: with supply-side TFP as the dominant recurring disturbance, output expansions coincide with falling inflation and falling policy rates. Adding demand-side disturbances (investment-specific or preference shocks) would improve the nominal comovements, but at the cost of a less transparent stochastic environment; since the transition analysis conditions on the carbon-tax impulse rather than on unconditional comovements, I keep the parsimonious specification and flag these limitations explicitly. Because the financial accelerator is central to the policy results, Table 5 isolates its contribution to these moments.

Which shocks drive the fit. By construction, all reported moments are generated by the TFP and monetary-policy shocks alone: the carbon-tax innovations are excluded from the validation simulations, as they constitute the policy experiments of Section 5 rather than recurring business-cycle disturbances. The business-cycle properties of the model are therefore inherited entirely from the standard TFP–monetary block.

The role of financial frictions in the model's moments. Column 3 of Table 5 reports the same moments for the frictionless variant of the model used in Appendix A.3 (credit spreads and portfolio adjustment costs removed). The result is clear-cut: the unconditional business-cycle moments are essentially unchanged—the output autocorrelation moves from 0.91 to 0.89, consumption from 0.88 to 0.89, investment from 0.94 to 0.91, and the relative volatilities are similarly close (Appendix C.1)—with every difference small relative to the sampling bands. The model's persistence properties therefore originate in the real block—habit formation ($h = 0.7$) and investment adjustment costs ($\kappa_I = 5.5$)—rather than in the financial accelerator, whose role is to shape the *conditional* transmission of the transition shock rather than the unconditional moments. The concern that the financial friction might mechanically drive the welfare results through excess amplification is addressed directly by the policy analysis: Section 5 re-runs the full rule comparison in the frictionless economy and shows that the *ranking* of policy rules is unchanged—only the magnitude of the gains from accommodation shrinks. The financial accelerator amplifies the stakes of the trade-off; it does not create it.

Table 5: Model Moments Across Variants: AR(1) Coefficients

Variable	Data	Baseline	No fin. frictions
Output	0.85	0.91	0.89
Consumption	0.85	0.88	0.89
Investment	0.82	0.94	0.91
Hours	0.92	0.55	0.60
Inflation	0.50	0.70	0.69
Interest rate	0.97	0.84	0.84

Note: "No fin. frictions" removes credit spreads and portfolio adjustment costs as in Appendix A.3. Both variants use the TFP and monetary shocks only (the carbon-tax innovations are excluded from the validation by design). Relative standard deviations are reported in Appendix C.1.

4 Computational Strategy and Optimal Policy Search

To identify the optimal monetary policy response to the carbon transition, I employ a two-stage computational strategy that bridges standard linear-quadratic stabilization methods with non-linear welfare analysis. This approach allows me to robustly characterize the trade-offs between macroeconomic stability, emissions reduction, and household welfare under different Taylor rule specifications.

4.1 Step 1: Optimal Simple Rules (OSR)

In the first stage, I establish a baseline by computing the Optimal Simple Rule (Schmitt-Grohé and Uribe, 2007). This method relies on a first-order approximation of the model dynamics and minimizes a weighted sum of unconditional variances. I consider the model loss functions to characterize the central bank's objective derived in Debortoli et al. (2019):

$$L_{CB} = \mathrm{Var}(\pi_t) + 1.042 \cdot \mathrm{Var}(y_{gap,t}) + 0.077 \cdot \mathrm{Var}(dr_t)$$

I optimize the coefficients of the standard Taylor rule (ϕ_π, ϕ_y) and the emissions-augmented rule $(\phi_\pi, \phi_y, \phi_e)$. This linear optimization provides a convex initialization point for the subsequent non-linear analysis.

4.2 Step 2: Generalized Impulse Response Optimization

Linear approximations may fail to capture the full welfare implications of the transition, particularly given the presence of Epstein-Zin preferences and the non-linearities introduced by financial frictions and abatement technology. Therefore, in the second stage, I solve the model using a second-order perturbation with pruning, which is a suitable approach for analyzing welfare in a dynamic setting (Schmitt-Grohé and Uribe, 2004). I then perform a grid search over the policy parameter space, centered on the OSR results, to identify the rule that optimizes performance conditional on the specific carbon tax shock. For each parameter combination in the grid $(\phi_\pi, \phi_y, \phi_e)$, I compute Generalized Impulse Response Functions (GIRFs) by simulating the economy with and without the carbon tax shock over 2000 Monte Carlo replications. This effectively marginalizes out the influence of concurrent stochastic volatility. I evaluate each candidate rule according to the three criteria of Section 3.10: household welfare in consumption-equivalent units, the cumulative central-bank loss, and cumulative emissions.

This dual approach ensures that the identified optimal rules are robust to both general business cycle fluctuations (via OSR) and the specific structural break induced by the carbon transition (via GIRF).

5 Results: Optimal Monetary Policy under Transition Shocks

This section presents the main findings from the optimal monetary policy analysis during a carbon tax transition. I compare a standard Taylor rule, in which policy responds only to inflation and the output gap, with an emissions-augmented rule that additionally responds to emissions dynamics. The analysis evaluates both rules under two objectives: maximizing household welfare, measured in consumption-equivalent terms, and minimizing a conventional quadratic central bank loss function. Results are organized around the three carbon-tax scenarios of Section 3.9: the permanent unanticipated increase (the reference scenario, Table 6) and the two transition-path variants—the persistent surprise and the anticipated phase-in (Table 7), with Table 8 summarizing the gains and losses from augmenting the policy rule.

5.1 The Reference Scenario: A Permanent Carbon Tax

Table 6 reports optimal policy coefficients and outcomes when the carbon tax increase is permanent and unanticipated ($\rho_\tau = 0.9999$), the reference scenario of the analysis. Because the tax never reverts, any accommodation motive that relied on an anticipated reversal of the policy is absent by construction.

Table 6: Optimal Taylor Rule Parameters and Outcomes: Permanent Carbon Tax

Criterion	ϕ_π	ϕ_y	ϕ_e	CE (%)	$L_{CB} \times 100$	e_{cum}
Std (Max Welf)	3.00[a]	0.15	—	−0.343	0.142	−4.519
Std (Min L_{CB})	1.10	0.05	—	−0.350	0.105	−4.510
Aug (Max Welf)	1.40	0.00	0.100	−0.306	0.574	−4.369
Aug (Min L_{CB})	1.30	0.05	0.025	−0.331	0.075	−4.484

Notes: Permanent unanticipated carbon tax ($\rho_\tau = 0.9999$), implemented as a single unanticipated impulse with the recurring carbon-tax shock variance switched off (TFP and monetary shocks retained as background risk). Grid and GIRF settings as in Table 7. [a]Corner of the search grid; the welfare surface of the standard rule is nearly flat in ϕ_π under the permanent tax (CE ranges only from −0.343 to −0.350 across the reported standard rules).

The central results survive the removal of mean reversion, and the stakes become larger. Under the stabilization objective, the optimal emissions response is *identical* to the transitory-tax scenarios ($\phi_e^* = 0.025$), and under the welfare objective the accommodation motive persists at the same magnitude ($\phi_e^* = 0.100$). Two features of the model explain this invariance. First, the augmented rule responds to emissions *growth*, not to deviations from the initial steady state; the informational content of emissions momentum is the same whether the economy transits to a new equilibrium or eventually returns to the old one. Second, the welfare gains from accommodation stem from smoothing the *adjustment*—the front-loaded contraction of brown asset values and bank net worth—which occurs on impact regardless of the long-run level of the tax. Quantitatively, permanence raises the welfare cost of the transition roughly threefold (CE of −0.31 to −0.35 percent, against −0.08 to −0.10 under the persistent surprise) and *increases* the value of the emissions signal: augmenting the loss-minimizing rule now reduces the discounted central-bank loss by 29 percent (against 4–5 percent under the transitory tax) and raises welfare by 0.019 percentage points, at the cost of a 0.6 percent smaller cumulative emissions decline. The welfare gain from aggressive accommodation is likewise larger (0.037 against 0.024 percentage points). The remainder of this section analyzes the two transition-path variants—imperfect credibility and legislated gradualism—which preserve the ranking of rules while muting the quantitative stakes; impulse responses for the permanent scenario, including the financial variables, are reported in Appendix A.6.

Table 7: Optimal Taylor Rule Parameters and Outcomes: Standard ($\phi_e = 0$) vs Emissions-Augmented (ϕ_e free)

Method	Criterion	ϕ_π	ϕ_y	ϕ_e	CE (%)	$L_{CB} \times 100$	e_{cum}
Panel A: OSR (Order=1, Unconditional Variance)							
OSR (L_{CB})	Standard	1.14	0.11	—	—	Var=0.26	—
OSR (L_{CB})	Augmented	1.15	0.11	0.014	—	Var=0.23	—
Panel B: GIRF (Order=2, Surprise Shock $\varepsilon_\tau^{surprise}$)							
GIRF	Std (Max Welf)	1.10	0.15	0.000	-0.099	0.047	-4.270
GIRF	Std (Min L_{CB})	1.10	0.10	0.000	-0.100	0.044	-4.275
GIRF	Aug (Max Welf)	1.50	0.12	0.100	-0.075	0.303	-4.200
GIRF	Aug (Min L_{CB})	1.50	0.25	0.025	-0.096	0.042	-4.266
Panel C: GIRF (Order=2, Gradual Shock $\varepsilon_\tau^{gradual}$)							
GIRF	Std (Max Welf)	1.10	0.15	0.000	-0.100	0.049	-3.687
GIRF	Std (Min L_{CB})	1.10	0.10	0.000	-0.100	0.045	-3.693
GIRF	Aug (Max Welf)	1.10	0.30	0.075	-0.073	0.187	-3.625
GIRF	Aug (Min L_{CB})	1.40	0.15	0.025	-0.097	0.043	-3.684

Notes: Standard Taylor rules restrict $\phi_e = 0$. Augmented rules allow ϕ_e to vary. Columns report: (1) Welfare as Consumption Equivalent (CE %), (2) Cumulative Discounted CB Loss (L_{CB}), and (3) Cumulative Discounted Emissions (e_{cum}). GIRF metrics computed under order=2 pruned simulation.

5.2 Transition Paths: Persistent Surprise and Anticipated Phase-In

5.3 Optimal Policy Coefficients

Across the two mean-reverting transition paths, the standard Taylor rule delivers a strikingly consistent pattern. When the emissions coefficient is constrained to zero, the optimal inflation response lies close to the lower bound of the admissible range, with ϕ_π between 1.10 and 1.14, and the output-gap coefficient between 0.10 and 0.15. This configuration reflects the supply-side nature of the carbon tax shock. Aggressive inflation stabilization would exacerbate the output contraction induced by higher production costs, while a more accommodative stance allows inflation to absorb part of the adjustment. As a result, the standard rule effectively accommodates the transition, but at the cost of non-trivial welfare losses of about 0.10 percent in consumption-equivalent terms. The permanent benchmark behaves differently on this margin: as noted in Section 5.1, the welfare surface of the standard rule is nearly flat in ϕ_π there (reaching the upper grid boundary), because with a never-reverting tax the inflation coefficient has little leverage over a supply adjustment that must happen regardless; the loss-minimizing standard rule, by contrast, remains accommodative ($\phi_\pi = 1.10$) under all three scenarios.

Allowing the emissions coefficient to vary alters the optimal policy design in a systematic way. In all cases, the optimized emissions response is positive, ranging from a small value under the first-order OSR solution to substantially larger values under second-order GIRF optimization when welfare is the objective. A positive emissions coefficient implies that policy eases as emissions fall during the transition, providing additional demand support when carbon pricing tightens firms' cost conditions. At the same time, the augmented rules are paired with higher inflation coefficients, often in the range of 1.40 to 1.50. This combination indicates that emissions accommodation and inflation vigilance act as complements rather than substitutes: explicit recognition of the transition shock allows monetary policy to lean more aggressively against inflation without inducing excessive real contraction.

5.4 Welfare and Stabilization Trade-offs

Table 8 reports the trade-offs associated with augmenting the policy rule with emissions. When welfare is the objective, the augmented rule delivers positive but modest gains. Relative to the standard welfare-optimal rule, consumption-equivalent welfare increases by 0.024 percentage points following a surprise carbon tax shock and by 0.026 percentage points under an anticipated (gradual) shock. The larger gain in the latter case reflects the interaction between forward-looking adjustment and monetary accommodation.[6]

Table 8: Gains/Losses of Emissions-Augmented Rule Relative to Standard Rule Benchmarks ($\phi_e = 0$)

	Surprise Shock			Anticipated Shock		
Comparison	ΔCE (pp)	ΔL_{CB} (%)	Δe (%)	ΔCE (pp)	ΔL_{CB} (%)	Δe (%)
Aug (Max Welfare) vs Std (Max Welfare)	0.024	-549.69	1.6	0.026	-280.46	1.7
Aug (Min L_{CB}) vs Std (Min L_{CB})	0.004	4.38	0.2	0.003	4.71	0.2

Notes: ΔCE is the difference in welfare (consumption-equivalent) in percentage points. ΔL_{CB} is the percent reduction in discounted CB loss. Δe is the percent reduction in cumulative emissions.

The mechanism behind these gains is intertemporal reallocation. A positive emissions coefficient implies that policy eases as emissions fall during the transition, partially offsetting the contractionary effects of higher production costs and improving the average consumption and leisure paths during

[6] To contextualize the magnitude of these welfare gains, Schmitt-Grohé and Uribe (2007) find that moving from sub-optimal to welfare-maximizing simple rules yields gains of 0.01–0.05 percentage points in similar DSGE frameworks. The estimates of 0.02–0.04 percentage points fall squarely within this range, suggesting the gains from emissions augmentation are economically meaningful, comparable to removing moderate macroeconomic volatility.

the adjustment. The gains are nonetheless modest, because the same accommodation raises the dispersion of consumption across states of the world, and the recursive preferences—which place a high premium on short-run consumption risk—price this added dispersion against the mean-path improvement; the decomposition in Section 5.5 quantifies both sides of this trade-off. The gains reflect a smoother average transition rather than higher long-run consumption.

The welfare improvements under aggressive emissions accommodation are obtained at the expense of traditional stabilization performance. In both shock environments, the welfare-maximizing augmented rule raises the discounted central bank loss sharply relative to the standard rule, with L_{CB} worsening by roughly 280 to 550 percent. This deterioration reflects higher inflation and output-gap volatility because monetary policy is accommodating a cost-push disturbance rather than leaning against it. From the perspective of an institution primarily charged with price stability, this policy would be difficult to reconcile with a traditional mandate.

When the objective is to minimize L_{CB}, the optimal augmented rule becomes substantially more conservative. The emissions response remains positive but small, and the rule places greater weight on inflation stabilization. In this case, augmentation modestly improves welfare, with ΔCE between 0.003 and 0.004 percentage points, while reducing the central bank loss by about 4 to 5 percent. The implied environmental differences are negligible, with cumulative emissions changing by only about 0.2 percent relative to the standard loss-minimizing benchmark. Thus, a limited emissions response can yield small welfare improvements and better stabilization outcomes without altering the emissions path. In practice, this implies that the central bank should not attempt to actively drive the climate transition, but can treat emissions dynamics as an auxiliary indicator of supply-side pressures, using them to fine-tune policy without compromising price stability or environmental policy objectives.

These results do not imply that optimal monetary policy should disregard inflationary pressures associated with the green transition. On the contrary, inflation plays a central role in the transmission of the carbon tax shock across all policy regimes. The welfare-maximizing augmented rule accommodates inflation temporarily as a deliberate mechanism to smooth real adjustment costs in the presence of a supply-driven price increase, not because inflation stabilization is unimportant. This accommodation comes at the cost of large and persistent deviations of inflation from target, which explains the sharp deterioration in the central bank loss under welfare maximization. When the policy objective instead reflects a conventional stabilization mandate, the optimal response tightens inflation control substantially, and the emissions coefficient becomes small. In this case, inflation remains well contained, nominal volatility is limited, and welfare gains from emissions augmentation are correspondingly modest. The contrast between these regimes stresses that greenflation is not neglected in the analysis, but rather explicitly traded off against real stabilization, with mandate-consistent policies placing primary weight on price stability.

5.5 Decomposing the Welfare Effects

To clarify the economic mechanisms behind the welfare numbers, Table 9 decomposes the consumption-equivalent effect of the transition under each rule into four components: the mean consumption path, the hours/leisure response, consumption risk (the component priced by the recursive preferences), and the environmental component operating through the damage function.

Table 9: Decomposition of Consumption-Equivalent Welfare Effects (pp)

Rule	Total CE	Mean cons.	Leisure	Cons. risk	Environment
Std (Max Welf)	−0.099	−0.283	+0.081	+0.102	+0.001
Std (Min L_{CB})	−0.100	−0.294	+0.085	+0.109	+0.001
Aug (Max Welf)	−0.081	−0.079	+0.175	−0.177	+0.001
Aug (Min L_{CB})	−0.096	−0.273	+0.103	+0.074	+0.001

Notes: Surprise carbon tax shock. "Mean cons." is the CE effect of the average consumption path across replications; "Leisure" the contribution of the hours response; "Cons. risk" the contribution of dispersion across stochastic paths, obtained as the difference between the full stochastic CE and the mean-path CE; "Environment" converts the damage-function implications of the emissions path into consumption units. Components sum to the total up to a small approximation residual. Computation described in Appendix C.2.

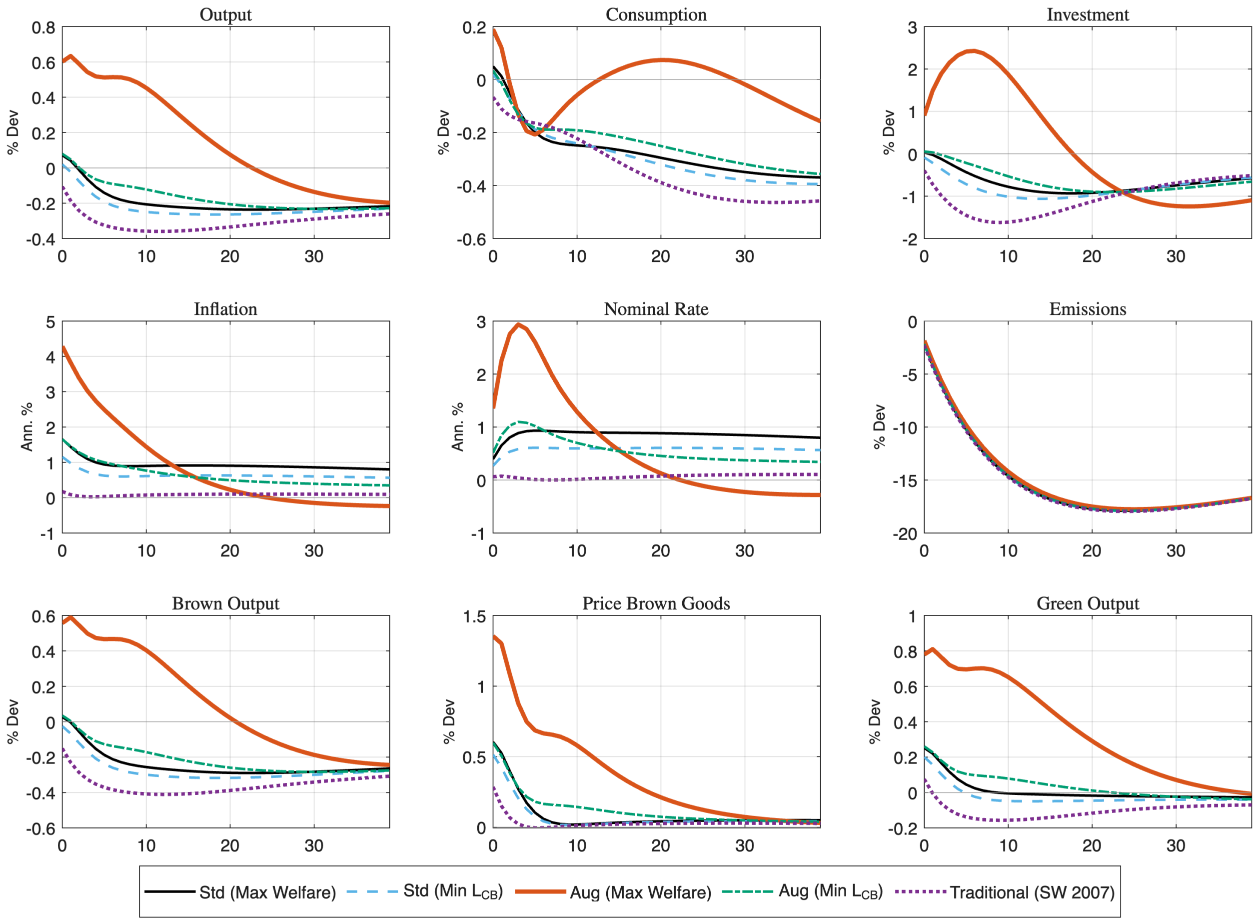
The decomposition reveals that aggressive accommodation works by *relocating* the welfare cost of the transition rather than eliminating it. Under the standard rules, virtually the entire loss is borne by the mean consumption path (−0.28 to −0.29 percentage points), partially offset by the leisure gain from lower hours. The welfare-maximizing augmented rule nearly eliminates the mean-consumption loss (−0.08) and doubles the leisure gain—this is the investment boom at work—but in doing so it *reverses the sign* of the consumption-risk component, from +0.10 to −0.18: the boom-bust dynamics it engineers make consumption paths substantially more dispersed across states of the world, and recursive preferences price this added risk heavily. The result is that most of the mean-path improvement is clawed back by the risk penalty, which explains why the net welfare gain of aggressive accommodation is modest (0.02–0.03 percentage points) despite its dramatic effect on average quantities. The loss-minimizing augmented rule, by contrast, obtains a smaller mean-path improvement while *preserving* a positive risk component—accommodating enough to smooth the adjustment without generating boom-bust dispersion. The environmental component is negligible for all rules, confirming that monetary policy operates on the macroeconomic adjustment, not on climate outcomes. This decomposition sharpens the paper's central message: the relevant margin for evaluating transition-time monetary policy is the trade-off between the average adjustment path and the riskiness of that path—a margin that a quadratic central-bank loss function does not capture.

5.6 Dynamic Adjustment and Transmission

The impulse response functions in Figures 1 and 2 illustrate the distinct transmission mechanisms across the five policy regimes. The inclusion of a traditional Taylor rule benchmark ($\phi_\pi = 2.0, \phi_y = 0.08$) (Smets and Wouters, 2007) emphasizes the role of monetary strictness in shaping the transition. Under this strict inflation-targeting regime, the carbon tax acts as a pure supply shock: the central bank raises nominal rates to combat cost-push inflation, driving real interest rates higher. This precipitates a contractionary adjustment where investment, consumption, and output contracts. This trajectory serves as the baseline for a transition driven purely by carbon pricing without monetary accommodation. In contrast, the optimal rules reveal a spectrum of accommodation.

The standard Taylor rules and the loss-minimizing augmented rule effectively mimic the traditional benchmark but with slightly dampened recessionary effects. By tolerating a modest increase in inflation, these rules avoid the sharpest contraction in output seen under the strict traditional rule. However, the overall dynamic remains recessionary: investment stagnates, and the transition proceeds through a contraction in aggregate demand that facilitates the reallocation of resources away from the brown sector. A different adjustment path emerges under the welfare-maximizing augmented rule (solid orange). By responding positively to the decline in emissions ($\phi_e > 0$), this policy effectively lowers the real interest rate despite rising inflation. This monetary accommodation triggers a massive, front-loaded expansion in investment and a temporary boom in output. Consumption falls due to intertemporal substitution, but the drop is less pronounced than under the other rules. This "Green Boom" comes at a steep nominal cost: inflation spikes to over 4% (annualized) and nominal rates exhibit extreme volatility. While theoretically optimal for household welfare in this model, such aggressive accommodation generates nominal instability that would likely be viewed as unfeasible or outside the mandate of a conventional inflation-targeting central bank.

Figure 1: Impulse Responses to Surprise Carbon Tax Shock

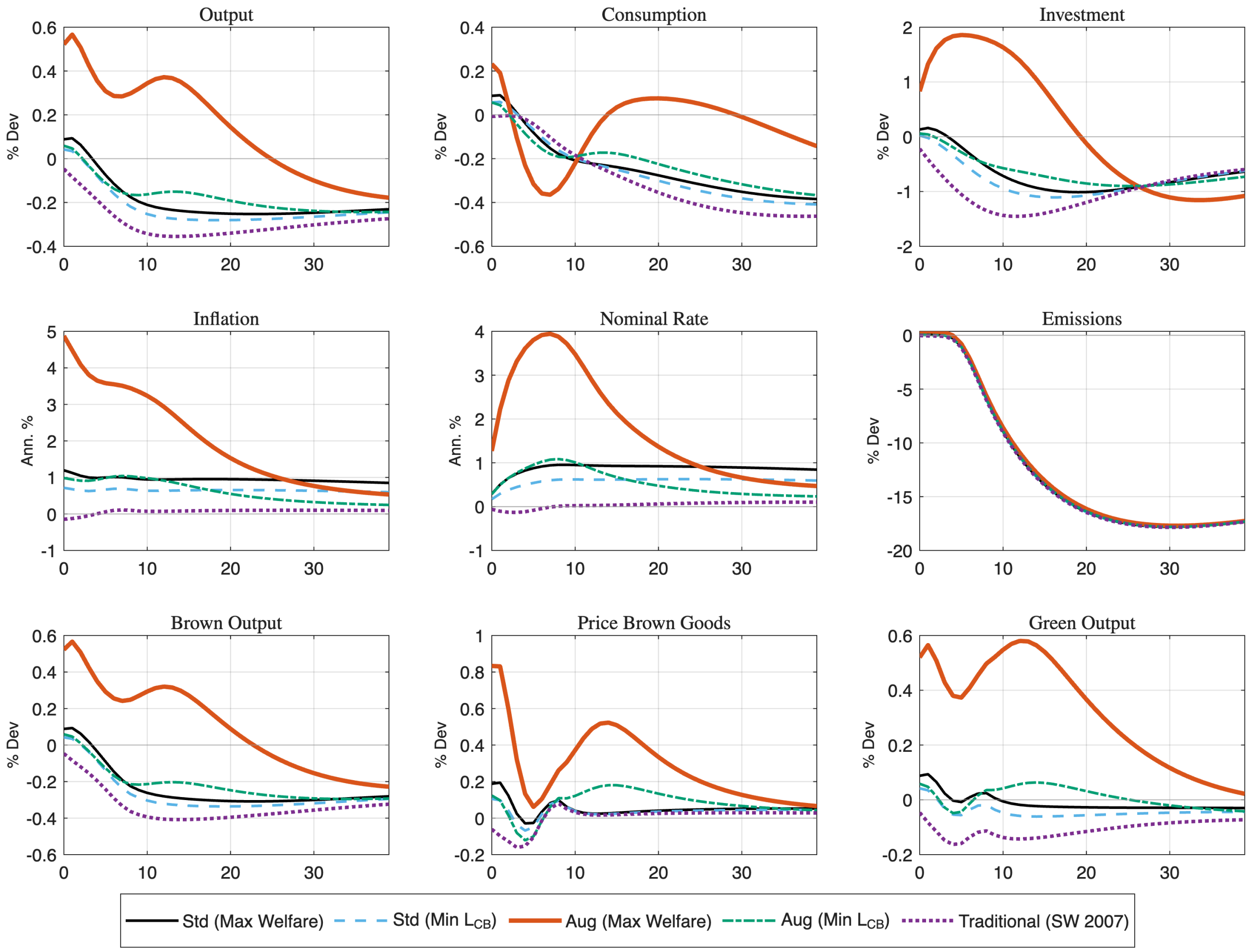


Impulse responses to a surprise increase in the carbon tax under five policy rules. Output, consumption, investment, emissions, brown output, and green output are expressed as percent deviations from baseline. Inflation and nominal rate are expressed in annualized percentage points. Horizontal axis shows quarters after the shock.

The financial transmission channel behind these divergent paths—a "Financial Crunch" under strict rules versus "Financial Shielding" under aggressive accommodation—is analyzed in detail in Section 5.7.

Comparing shock structures, the anticipation of the tax (Figure 2) smooths the adjustment but does not alter the ranking of policies. The gradual phase-in allows forward-looking agents to adjust capital stocks earlier, mitigating the initial drop in consumption. However, the fundamental trade-off remains: strict adherence to price stability (Traditional/Loss-Minimizing) enforces a recessionary transition, while aggressive emissions-targeting (Welfare-Maximizing) can engineer an expansionary transition by tolerating high inflation volatility.

Figure 2: Impulse Responses to Gradual (Anticipated) Carbon Tax Shock



Impulse responses to a gradual carbon tax increase announced at $t = 0$ and phased in over quarters 4–8. Policy rules and variable definitions as in Figure 1. Horizontal axis shows quarters after the shock.

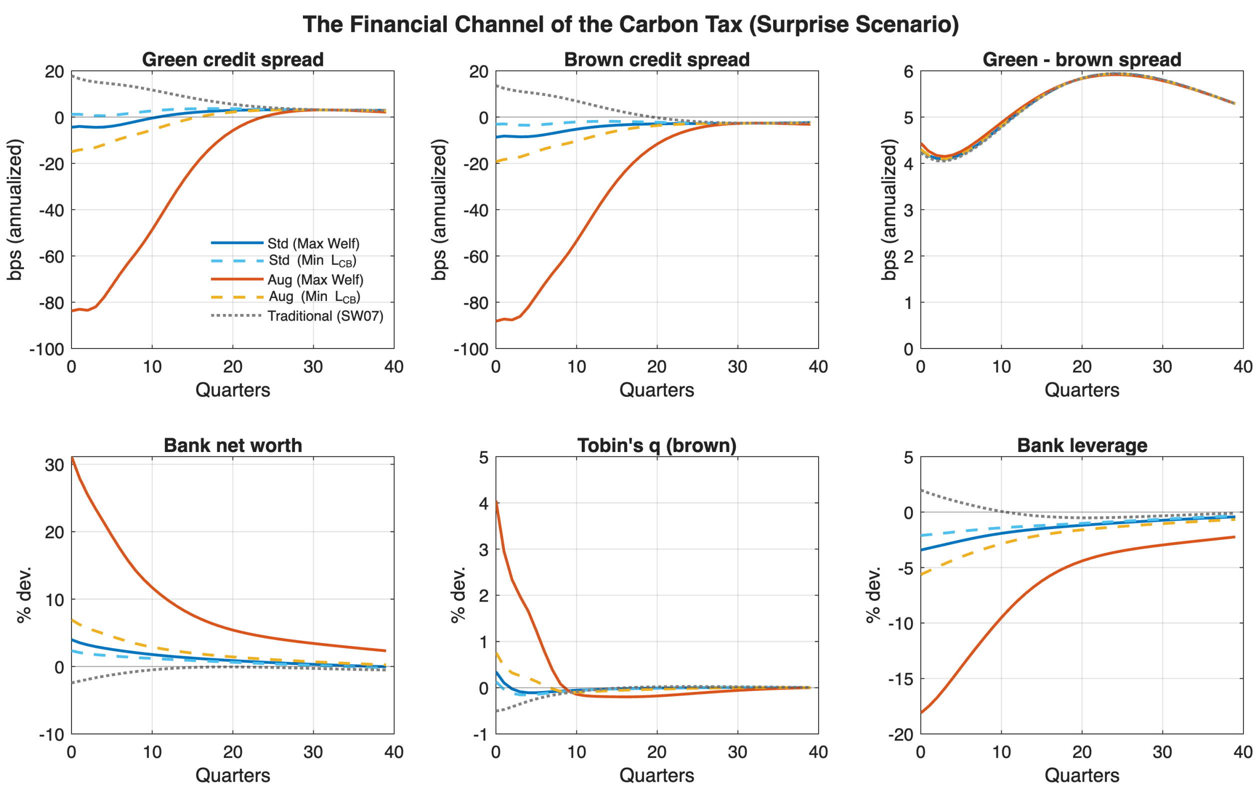
These impulse responses indicate that emissions-augmented monetary policy primarily reallocates adjustment costs over time rather than altering the long-run emissions path, which is largely determined by the carbon tax. Aggressive augmentation operates by accommodating the transition through an inflationary investment boom, shifting volatility away from real activity toward nominal variables. The ranking of policy rules is stable across surprise and anticipated shocks, suggesting that the central bank's optimal response depends more on its tolerance for nominal volatility than on the timing of the transition. The inflation dynamics under welfare maximization should therefore be interpreted as an upper bound on feasible accommodation, rather than as a realistic policy path.

The expansionary response observed under aggressive accommodation reflects the dominance of the monetary policy channel over the carbon tax's direct cost-push effects. While the tax itself is contractionary, the strong easing induced by falling emissions more than offsets this effect, underscoring that the macroeconomic impact of carbon policy is highly sensitive to the monetary regime under which it is implemented.

5.7 The Financial Channel: Credit Spreads and Bank Balance Sheets

Figure 3 traces the financial transmission of the carbon tax under the five policy rules: green and brown credit spreads, the green–brown spread differential, bank net worth, Tobin's q in the brown sector, and bank leverage.

Figure 3: Financial Channel of the Carbon Tax Shock



Impulse responses of sectoral credit spreads (annualized basis points), bank net worth, brown-sector Tobin's q, and bank leverage to a surprise carbon tax increase under five policy rules. Spreads are defined as the expected excess return of sectoral bonds over the risk-free rate.

Under the traditional inflation-targeting rule, the carbon tax triggers the classic financial accelerator: bank net worth contracts by 2.4 percent and both credit spreads widen—the green spread by 18 and the brown spread by 14 basis points at peak. Two features of this response are noteworthy.

First, the tax transmits sectoral stress to the *aggregate* credit supply: because both sectors borrow from the same intermediaries, green firms' financing conditions tighten even though the tax falls on brown production. Second, the widening is asymmetric *against* green credit: as the value of brown bonds falls, banks' portfolio share of green bonds rises above its target, and the portfolio friction then commands a premium for holding additional green exposure. Green producers thus face higher funding costs through two channels—the contraction of aggregate bank net worth and the portfolio-rebalancing premium—which constitutes the model's key financial externality of carbon taxation.

Monetary policy determines whether this crunch materializes. The optimized standard rules already mute it almost entirely (peak spread movements within ± 9 basis points; net worth essentially flat). The welfare-maximizing augmented rule inverts it: by easing as emissions fall it supports asset prices, net worth *rises* by 2.3 percent, and spreads compress by 84–88 basis points—the "financial shielding" that finances the investment boom described above. The loss-minimizing augmented rule sits in between, compressing spreads by 15–19 basis points at negligible inflationary cost. Table 10 collects the peak responses. In sharp contrast to the level of spreads, the *green–brown differential* peaks at +5.9 basis points under every one of the five rules: monetary policy has no traction on the relative price of green versus brown credit, which is pinned down by the carbon tax and the portfolio friction. What the central bank controls is the aggregate tightness of credit during the transition—a precise financial-sector counterpart of the paper's division-of-labor result between monetary and fiscal instruments.

Table 10: Peak Responses of Financial Variables: Surprise Carbon Tax Shock

Rule	Peak sp^G (bps)	Peak sp^B (bps)	Peak $sp^G - sp^B$ (bps)	Trough NW (%)
Std (Max Welf)	−4.4	−8.7	+5.9	−0.03
Std (Min L_{CB})	+3.7	−3.5	+5.9	−0.15
Aug (Max Welf)	−83.8	−88.2	+5.9	+2.33
Aug (Min L_{CB})	−14.9	−19.2	+5.9	+0.25
Traditional (SW07)	+17.8	+13.6	+5.9	−2.43

Notes: Peak = largest absolute deviation from baseline over the 40-quarter window, with sign; spreads in annualized basis points. Trough NW is the extreme response of bank net worth in percent (positive entries indicate that net worth rises throughout). Corresponding statistics for the permanent benchmark are visible in Appendix A.6.

5.8 The Role of Financial Frictions: A Frictionless Comparison

To assess the role of financial frictions in shaping the transition dynamics, Appendix A.3 reports impulse responses under an alternative specification in which banking frictions are shut down by eliminating credit spreads and leverage adjustment costs. In this frictionless environment, the qualitative ranking of policy rules is preserved, but the transmission mechanism and the quantitative differences across rules change substantially.

The key difference is in investment dynamics. Under the welfare-maximizing augmented rule and a surprise carbon tax, peak investment rises by approximately 5 percent in the baseline model with financial frictions, compared to less than 2 percent in the frictionless economy. This nearly three-fold amplification reflects the financial accelerator at work: by lowering real interest rates, the augmented rule supports Tobin's Q, strengthens bank net worth, and compresses credit spreads, generating a feedback loop between asset valuations and lending capacity. Without this loop, monetary easing still stimulates investment through standard intertemporal substitution, but the response is substantially weaker. Output responses are correspondingly attenuated, though the difference is more moderate than for investment.

Interestingly, the attenuation is not uniform across variables. Consumption under the welfare-maximizing rule is somewhat larger in the frictionless economy, as the absence of intermediation frictions allows more of the monetary easing to pass through to household spending rather than being absorbed by bank balance-sheet repair. Similarly, the inflation response is not attenuated, if anything, it is marginally larger without financial frictions. This asymmetry suggests that financial frictions redirect the effects of monetary accommodation toward investment rather than prices, rather than simply amplifying all responses uniformly.

Emissions dynamics are nearly identical across the two environments, confirming that the transition path is driven primarily by the carbon tax and abatement technology rather than by financial conditions. Monetary policy affects emissions only indirectly through aggregate activity, and the similarity of emissions paths reinforces the interpretation that the augmented rule responds to transition-induced macroeconomic stress rather than mechanically targeting environmental outcomes.

These results clarify the state-contingent nature of emissions-augmented monetary policy. The value of augmentation is greatest when transition shocks interact with financial frictions to generate balance-sheet stress and credit tightening—precisely the conditions under which conventional stabilization rules leave the most welfare on the table. In the frictionless economy, the gap between welfare-maximizing and loss-minimizing policies narrows substantially: without the financial accelerator, aggressive accommodation delivers smaller real gains while still generating comparable inflation volatility, making the welfare-stabilization trade-off less acute. Financial frictions are therefore important not only to the strength of monetary transmission during the transition, but also to the practical tension between welfare-oriented and mandate-consistent policy design.

5.9 Robustness checks - Alternative Parametrization

To assess whether the divergence between welfare-maximizing and stabilization-oriented monetary policy reflects a genuine economic trade-off rather than a feature of the baseline calibration, Table 11 reports optimal policy coefficients (ϕ_π^*, ϕ_e^*) across a broad set of alternative specifications for the surprise shock (Appendix A.1 presents the same table for the gradual shock). The robustness analysis spans variation in household preferences, structural rigidities, production technologies, and the size of the carbon tax shock. A note on comparability: the robustness batteries (Tables 11–15) use 1,000–2,000 order-2 GIRF replications with a 200-quarter burn-in, whereas the headline optimizations (Tables 6–8) use 2,000 replications with a 500-quarter burn-in; welfare *levels* are therefore comparable within each table but can differ by up to 0.02 percentage points across tables, while optimal coefficients and rule rankings are unaffected by these settings. Each table states its own settings and its own baseline row.

The main conclusion is that the qualitative policy trade-off identified in the baseline is highly stable. Across changes in risk aversion (σ), intertemporal elasticity of substitution (ψ), habit persistence (h), and the elasticity of substitution across inputs (ξ), the ranking of policy regimes remains essentially unchanged. When welfare is the objective, the optimal rule consistently features a sizable positive response to emissions, with ϕ_e^* clustered around 0.10. In this regime, monetary policy exploits the financial accelerator to generate an investment-driven expansion that cushions the real effects of the transition. By contrast, when the objective is to minimize the central bank loss, the optimal emissions response remains small, typically around $\phi_e^* \approx 0.025$, reflecting a strong preference for limiting nominal volatility. These patterns indicate that the tension between real smoothing and price stability is structural rather than parameter-driven.

Table 11: Robustness of Optimal Taylor Rule Parameters

	Max Welfare			Min L_{CB}		
Parameter	ϕ_π^*	ϕ_e^*	CE (%)	ϕ_π^*	ϕ_e^*	L_{CB}
Risk Aversion (σ)						
$\sigma = 2$	1.50	0.100	-0.089	1.10	0.025	0.0952
$\sigma = 10$ (baseline)	1.50	0.100	-0.084	1.10	0.025	0.0952
$\sigma = 30$	1.50	0.075	-0.091	1.10	0.025	0.0952
Intertemporal Elasticity of Substitution (ψ)						
$\psi = 0.5$	1.50	0.100	-0.092	1.10	0.025	0.108
$\psi = 1.5$ (baseline)	1.50	0.100	-0.084	1.10	0.025	0.0952
$\psi = 2.0$	1.50	0.100	-0.089	1.10	0.025	0.0936
$\psi = 5.0$	1.50	0.100	-0.088	1.10	0.025	0.0911
Carbon Tax Shock Size						
0.5%	1.50	0.200	-0.030	1.10	0.025	0.098
1.0% (baseline)	1.50	0.100	-0.084	1.10	0.025	0.091
2.0%	1.10	0.050	-0.220	1.10	0.025	0.093
5.0%	1.50	0.025	-0.762	1.50	0.025	0.099
10.0%	1.50	0.000	-2.127	1.50	0.025	0.102
Habit Persistence (h)						
$h = 0.0$	1.50	0.100	-0.087	1.10	0.025	0.097
$h = 0.3$	1.50	0.100	-0.087	1.10	0.025	0.096
$h = 0.7$ (baseline)	1.50	0.100	-0.084	1.10	0.025	0.095
$h = 0.8$	1.50	0.100	-0.090	1.10	0.025	0.095
Abatement Technology						
No Abatement	1.50	0.200	-0.067	1.50	0.200	0.196
With Abatement (baseline)	1.50	0.100	-0.084	1.10	0.025	0.095
CRRA Preferences ($\psi = 1/\sigma$)						
$\sigma = 2$	1.50	0.100	-0.092	1.10	0.025	0.108
$\sigma = 5$	1.50	0.075	-0.100	1.10	0.025	0.142
$\sigma = 10$	1.50	0.075	-0.109	1.50	0.025	0.194
Elasticity of Substitution (ξ)						
$\xi = 1.01$	1.50	0.100	-0.086	1.10	0.025	0.096
$\xi = 2.0$ (baseline)	1.50	0.100	-0.084	1.10	0.025	0.095

Notes: Optimal (ϕ_π, ϕ_e) selected over the grid $\phi_\pi \in \{1.10, 1.50\}$ and $\phi_e \in [-0.10, 0.15]$, holding $\phi_y = 0.125$ fixed. CE denotes consumption-equivalent welfare. L_{CB} is the cumulative discounted central bank loss. Results based on 2,000 order-2 GIRF replications with pruning.

The importance of technological constraints is illustrated by the no-abatement scenario. When firms lack access to abatement technologies and can only respond to carbon pricing by cutting output, the transition shock becomes significantly more contractionary. In this case, the optimal emissions coefficient rises sharply for both objectives, reaching $\phi_e^* = 0.20$. The central bank responds by providing aggressive financial accommodation to offset the lack of technological adjustment margins and prevent a collapse in aggregate demand. This result highlights that the scope for monetary stabilization expands when the real economy is less flexible, even though such policies may come at a higher inflationary cost.

The exercise also reveals a pronounced non-linearity with respect to the size of the carbon tax shock. For relatively small shocks, monetary policy can afford to respond aggressively to emissions dynamics, effectively smoothing the transition through accommodative financial conditions without destabilizing inflation. As the magnitude of the shock increases, however, the optimal emissions response declines rapidly and eventually converges toward zero. Large carbon tax increases generate substantial cost-push inflation on their own, leaving little room for additional accommodation. In such environments, a positive emissions response would amplify inflationary pressures to an extent that is inconsistent with price stability. This non-linearity underscores a fundamental limit to the role of monetary policy in managing the transition: when the shock itself threatens nominal stability, the central bank must revert to orthodox inflation control.

Severity of climate damages. The DICE damage function is a conservative benchmark, and recent estimates imply substantially larger macroeconomic costs of warming (Bilal and Känzig, 2026). Table 12 re-optimizes the policy rules with the damage coefficients $(\tilde{D}_1, \tilde{D}_2)$ scaled by factors of 0.5 to 4, the latter approximating damage levels consistent with these recent estimates. The optimal coefficients are remarkably stable: $\phi_e^* = 0.100$ under the welfare objective and $\phi_e^* = 0.025$ under the loss objective at every severity level. The reason is that damages operate through the slow-moving pollution stock, so even a fourfold increase in severity barely alters the 40-quarter stabilization problem; what changes is the *net* welfare cost of the transition, which falls monotonically with severity (CE from -0.103 at $0.5\times$ to -0.073 at $4\times$) because more severe damages raise the environmental payoff of the same tax-induced emissions reduction. Notably, the concern that harsher damages would penalize the emissions-delaying investment boom of the welfare-maximizing rule does not materialize at these horizons: the short-run emissions delay is too small relative to the stock dynamics to overturn the accommodation motive. Damages remain deterministic throughout; introducing climate-system uncertainty is left for future work.

Intensity of financial frictions and abatement costs. Table 13 varies the tightness of the banking friction (steady-state leverage, the credit-spread target, and the portfolio adjustment cost

Table 12: Robustness: Severity of the Damage Function

	Max Welfare			Min L_{CB}		
Damage scale	ϕ_π^*	ϕ_e^*	CE (%)	ϕ_π^*	ϕ_e^*	$L_{CB} \times 100$
0.5×	1.50	0.100	−0.103	1.50	0.025	0.0297
1× (baseline)	1.50	0.100	−0.099	1.50	0.025	0.0287
2×	1.50	0.100	−0.091	1.50	0.025	0.0268
4× (severe)	1.50	0.100	−0.073	1.50	0.025	0.0233

Note: Surprise carbon tax shock; $\phi_y = 0.125$ fixed; $\phi_\pi \in \{1.10, 1.50\}$ optimized jointly with ϕ_e; 1,000 order-2 GIRF replications.

κ_F) and the curvature of the abatement cost function. The policy prescription is highly stable: the loss-objective response is $\phi_e^* = 0.025$ in every variant, and the welfare-objective response is $\phi_e^* = 0.100$ in all leverage, spread, and κ_F variants except one (at the 150-basis-point spread target it edges down to 0.075, one grid step). The only parameter that moves the optimal coefficients economically is the abatement cost curvature: when abatement is more costly ($\theta_2 = 2.0$), the intensive margin shrinks and the optimal accommodation rises ($\phi_e^* = 0.150$ and 0.050 under the two objectives), mirroring the no-abatement case; when abatement is cheaper ($\theta_2 = 3.2$), the opposite occurs. One boundary of the sensitivity analysis is structural rather than numerical: the Gertler–Karadi steady state requires $\chi(sp \cdot \phi_{ss} + 1/\beta) < 1$ (a positive start-up transfer for entering bankers), which at the baseline $\chi = 0.965$ caps the product of the spread and leverage targets just above its calibrated value. The baseline thus sits near the feasibility frontier of the banking contract, and the robustness analysis accordingly explores looser, not tighter, friction calibrations.

Risk aversion, intertemporal substitution, and the timing of uncertainty resolution. A well-known property of Epstein–Zin preferences is that the configuration of risk aversion and intertemporal substitution determines the preferred timing of uncertainty resolution: agents prefer early resolution when $\sigma > 1/\psi$ and late resolution when $\sigma < 1/\psi$ (Epstein et al., 2014). The baseline calibration ($\sigma = 10$, $\psi = 1.5$) implies a strong preference for early resolution, which raises the demand for insurance against transition risk and could, in principle, hardwire the case for monetary accommodation. Table 14 maps the optimal emissions coefficient over a (σ, ψ) grid that includes late-resolution configurations.

The surface delivers a sharp answer. The loss-minimizing emissions response is $\phi_e^* = 0.025$ at *every* point of the surface, including all late-resolution configurations: the paper's practical policy

Table 13: Robustness: Financial Frictions and Abatement Costs

	Max Welfare			Min L_{CB}		
Variant	ϕ_π^*	ϕ_e^*	CE (%)	ϕ_π^*	ϕ_e^*	$L_{CB} \times 100$
Bank leverage target ϕ_{ss}						
$\phi_{ss} = 4$	1.50	0.100	-0.097	1.50	0.025	0.0287
$\phi_{ss} = 5$	1.50	0.100	-0.097	1.50	0.025	0.0284
$\phi_{ss} = 6$ (baseline)	1.50	0.100	-0.099	1.50	0.025	0.0287
Credit spread target						
100 bps	1.50	0.100	-0.096	1.50	0.025	0.0339
150 bps	1.50	0.075	-0.097	1.50	0.025	0.0302
200 bps (baseline)	1.50	0.100	-0.099	1.50	0.025	0.0287
Portfolio adjustment cost κ_F						
$\kappa_F = 0$	1.50	0.100	-0.099	1.50	0.025	0.0287
$\kappa_F = 3$ (baseline)	1.50	0.100	-0.099	1.50	0.025	0.0287
$\kappa_F = 6$	1.50	0.100	-0.099	1.50	0.025	0.0287
Abatement cost curvature θ_2						
$\theta_2 = 2.0$	1.50	0.150	-0.104	1.50	0.050	0.0440
$\theta_2 = 2.6$ (baseline)	1.50	0.100	-0.099	1.50	0.025	0.0287
$\theta_2 = 3.2$	1.10	0.050	-0.088	1.50	0.025	0.0204

Note: Surprise carbon tax shock; $\phi_y = 0.125$ fixed; $\phi_\pi \in \{1.10, 1.50\}$ optimized jointly with ϕ_e. Leverage and spread targets above the baseline violate the Gertler–Karadi feasibility condition $\chi(sp \cdot \phi_{ss} + 1/\beta) < 1$ and admit no steady state (see text).

Table 14: Optimal Emissions Response ϕ_e^* over (σ, ψ)

	Max Welfare				Min L_{CB}			
$\sigma \backslash \psi$	0.3	0.5	1.5	2.5	0.3	0.5	1.5	2.5
0.5	$0.075^\dagger$	$0.100^\dagger$	$0.100^\dagger$	0.100	$0.025^\dagger$	$0.025^\dagger$	$0.025^\dagger$	0.025
2	$0.075^\dagger$	$0.100^\dagger$	0.100	0.100	$0.025^\dagger$	$0.025^\dagger$	0.025	0.025
10 (baseline)	0.075	0.100	**0.100**	0.100	0.025	0.025	**0.025**	0.025
20	0.075	0.100	0.100	0.100	0.025	0.025	0.025	0.025

Notes: † indicates late-resolution configurations ($\sigma < 1/\psi$). $\phi_y = 0.125$ fixed; $\phi_\pi \in \{1.1, 1.5\}$ optimized jointly. Bold entries mark the baseline calibration.

prescription does not depend on the timing of uncertainty resolution embedded in the preference calibration. The welfare-maximizing response varies only along the ψ dimension—falling to 0.075 when the intertemporal elasticity is very low ($\psi = 0.3$) and settling at 0.100 for $\psi \geq 0.5$—and is entirely insensitive to risk aversion, holding ψ fixed. This confirms the mechanism developed next: what disciplines the scope for accommodation is households' willingness to substitute consumption intertemporally, not their aversion to risk or their preference for early resolution.

5.10 The Contribution of Recursive Preferences

Table 15 isolates the contribution of Epstein–Zin preferences by re-solving the model under expected utility (CRRA), which is nested at $\sigma = 1/\psi$. Two CRRA calibrations are considered: moderate risk aversion ($\sigma = 2$, $\psi = 0.5$) and risk aversion equal to the baseline EZ value ($\sigma = 10$, which forces $\psi = 0.1$). All other parameters, shocks, and rules are identical.

Table 15: Epstein–Zin versus CRRA Preferences: Macroeconomic and Welfare Effects

Preferences	Rule	CE (%)	$L_{CB} \times 100$	e_{cum}	Peak π	Peak I
Epstein–Zin ($\sigma = 10$, $\psi = 1.5$, baseline)						
	Std (Max Welf)	−0.127	0.075	−3.320	1.57	0.00
	Std (Min L_{CB})	−0.121	0.061	−3.324	1.28	−0.07
	Aug (Max Welf)	−0.099	0.266	−3.257	5.65	4.85
	Aug (Min L_{CB})	−0.113	0.036	−3.318	1.72	0.30
CRRA ($\sigma = 2$, $\psi = 0.5$)						
	Std (Max Welf)	−0.132	0.076	−3.319	1.73	−0.01
	Std (Min L_{CB})	−0.125	0.061	−3.324	1.39	−0.07
	Aug (Max Welf)	−0.103	0.239	−3.257	6.06	3.73
	Aug (Min L_{CB})	−0.116	0.034	−3.317	1.80	0.19
CRRA ($\sigma = 10$, $\psi = 0.1$)						
	Std (Max Welf)	−0.169	0.090	−3.318	2.30	−0.06
	Std (Min L_{CB})	−0.151	0.065	−3.322	1.75	−0.10
	Aug (Max Welf)	−0.125	0.184	−3.275	7.01	1.29
	Aug (Min L_{CB})	−0.131	0.033	−3.318	2.03	0.01

Notes: Surprise carbon tax shock; 1,000 order-2 GIRF replications. Peak π in annualized percentage points; Peak I in percent deviation. CRRA is nested in the recursive specification at $\sigma = 1/\psi$. Levels differ slightly from Table 7 because the robustness battery uses a shorter burn-in and fewer replications; comparisons *within* the table hold settings fixed. The gradual-shock version of this table (Table 17) and impulse responses for the CRRA variants are reported in Appendix A.5.

The comparison shows that the qualitative trade-off—accommodation improves welfare but destabilizes prices—survives under expected utility, while its quantitative anatomy changes materially. Under CRRA with $\sigma = 10$, the low implied intertemporal elasticity makes households reluctant to substitute consumption across time: the investment boom that characterizes the welfare-maximizing augmented rule largely collapses (peak investment of 1.3 percent, against 4.9 percent under Epstein–Zin), and the same monetary accommodation leaks into prices instead, with peak annualized inflation rising from 5.7 to 7.0 percent. Welfare losses are uniformly larger under CRRA, and the loss-minimizing policy places relatively more weight on inflation control. The choice of preference specification is therefore not merely technical—it determines the transmission mechanism: Epstein–Zin preferences (or, more precisely, a high intertemporal elasticity) activate a quantity-driven investment channel of accommodation, whereas low-IES expected utility activates a price-driven channel that worsens the stabilization trade-off. Appendix A.5 presents the corresponding impulse responses.

5.11 The Information Value of Emissions Data

The loss-minimizing results admit a useful reinterpretation: they quantify the *information value* of emissions data for a central bank with a conventional mandate. Moving from the best standard rule to the best augmented rule—holding the objective fixed at L_{CB}—reduces the discounted central-bank loss by 4–5% under the transitory-tax scenarios and by 29% under the permanent tax, while raising welfare by 0.003–0.019 percentage points and changing cumulative emissions by less than one percent. This is the value of observing emissions momentum as a signal of transition-related supply pressure that inflation and the output gap reveal only with a lag: because the carbon tax moves sectoral costs before its full pass-through to aggregate prices, emissions growth provides an early, mandate-consistent reading of the shock.

Two caveats qualify this number. First, official emissions statistics are published with delays and undergo substantial revisions, and independent estimates can disagree materially; measurement error in emissions would shrink the signal's value toward zero and argues for the small, cautious response coefficients found here rather than for aggressive emissions targeting. The augmented rule responds to a smoothed, lagged emissions signal (annual momentum with the most recent observation dated $t-1$), so it does not rely on contemporaneous emissions; it does not, however, impose a full publication delay, and imposing one—for instance, ending the signal window at $t-4$—would shrink the reported gains further, reinforcing the case for small coefficients. Second, the value of the signal is state-contingent: it is highest during an active carbon-price transition and negligible in tranquil periods, so the augmentation is best understood as a transition-specific refinement, not a permanent change to the reaction function.

5.12 Robustness - Monetary Policy Shocks

To assess whether emissions augmentation distorts standard monetary transmission, I examine the economy's response to an exogenous contractionary monetary policy shock under the standard, emissions-augmented, and traditional Taylor rules in Appendix A.4. Across all specifications, real responses are closely aligned. Output, consumption, and investment fall on impact and recover gradually, while inflation declines and the nominal interest rate rises temporarily. These dynamics are consistent with conventional New Keynesian transmission.

Allowing the policy rule to respond to emissions does not materially alter the real effects of monetary disturbances. Differences across rules are confined to nominal variables and reflect how the policy rule internalizes emissions dynamics rather than changes in the underlying demand-side transmission mechanism. In particular, emissions augmentation does not introduce amplification or attenuation of real activity in response to monetary shocks. This confirms that augmenting the policy rule with emissions preserves the standard stabilization properties of Taylor-rule-based monetary policy.

This result is important for interpretation: emissions augmentation affects the conduct of policy during transition shocks without compromising the central bank's ability to manage conventional monetary disturbances.

6 Discussion and Policy Implications

Before drawing policy lessons, a distinction between the positive and the normative content of the results is in order. The positive findings—that carbon taxation propagates through bank balance sheets, that financial frictions amplify the contraction, and that emissions dynamics contain advance information about transition-related supply pressure—follow from the model's mechanisms and are robust across the parametrizations examined. The normative statements, by contrast, are conditional on the model's structure and calibration: the welfare-maximizing rule is "optimal" only within the class of simple Taylor rules considered, under the specific preference specification, friction structure, and carbon-tax process assumed, and for a representative-agent economy that abstracts from distributional effects of both inflation and the transition. These results should therefore be read as characterizing the *direction and structure* of the trade-offs central banks face—not as an operational policy prescription. In particular, the finding that aggressive emissions accommodation maximizes welfare is best understood as a diagnostic of the tension between household welfare and mandate-consistent policy, and its external validity beyond the model rests on the robustness exercises of Section 5, which show that the qualitative ranking, but not the exact coefficients, transfers across specifications.

With this caveat in place, the results highlight a fundamental tension in the conduct of monetary policy during a green transition: the policy that maximizes household welfare is not compatible with the stabilization mandates of modern independent central banks. This tension is not quantitative but structural, shaping how emissions-augmented rules should be interpreted and implemented in practice.

The analysis focuses on the interest rate channel, which transmits transition shocks to the financial sector through asset valuations, collateral constraints, and bank net worth. While central banks also possess balance sheet tools that could target spreads more directly, existing evidence suggests that these instruments are limited by market depth and institutional constraints (Nakov and Thomas, 2026). By contrast, the financial shielding mechanism identified here shows that conventional interest rate policy can already relax financial conditions during the transition, conditional on the central bank's tolerance for inflation volatility.

6.1 The Infeasibility of the Welfare-Optimal Strategy

From a theoretical perspective, the welfare-maximizing augmented rule delivers the largest consumption-equivalent gains by engineering an investment boom that cushions the economy against the carbon tax. However, the mechanism required to achieve these gains renders the policy infeasible for three reasons.

First, the strategy relies on substantial inflation accommodation. As shown in Figure 1, the welfare-optimal rule generates inflation rates well above standard targets, pushing real interest rates into negative territory. While this supports capital accumulation, it directly conflicts with price stability mandates.

Second, the policy induces excessive nominal volatility. Relative to the standard rule, the welfare-maximizing strategy increases the central bank loss by several hundred percent (Table 8). Such volatility would likely undermine credibility and weaken the central bank's ability to stabilize future shocks.

Third, the resulting demand expansion partially offsets the emissions-reducing intent of the carbon tax. As reported in Table 7, cumulative emissions fall by less under the welfare-maximizing rule than under the standard benchmark. A policy that maximizes welfare through monetary expansion therefore risks diluting the effectiveness of fiscal climate policy.

6.2 A Pragmatic Approach: Emissions as a Supply-Side Indicator

In contrast, the loss-minimizing results provide a feasible, mandate-consistent policy benchmark. When the objective is stabilization, the optimal response to emissions is positive but small ($\phi_e \approx 0.025$). Emissions dynamics are best interpreted not as a policy target, but as an informative indicator of transition-related supply pressures.

Because emissions move closely with the contraction of brown production following the carbon tax, they provide timely information about the severity of the supply shock. A modest response to emissions allows the central bank to fine-tune the inflation–output trade-off without triggering the destabilizing boom-bust dynamics associated with welfare maximization. This pragmatic augmentation improves stabilization outcomes without materially affecting the emissions path determined by fiscal policy.

6.3 Monetary–Fiscal Interaction and Policy Coordination

Carbon taxation is a fiscal instrument, and the optimal monetary response characterized above is necessarily conditional on the assumed fiscal design. The scenario analysis speaks directly to this dependence, because the three carbon-tax paths can be read as three fiscal regimes: a fully credible permanent reform (reference scenario), an imperfectly credible reform that agents partially expect to be reversed (persistent surprise), and an announced, gradually implemented reform that is likewise not fully locked in (anticipated scenario). Across all three, the stabilization-oriented prescription is the same—a strong inflation response paired with a small, positive emissions coefficient ($\phi_e^* = 0.025$ in every scenario)—while the welfare-maximizing degree of accommodation and its costs vary with the fiscal design: anticipation magnifies both the welfare gains and the inflationary costs of accommodation, and permanence raises the welfare stakes of smoothing the adjustment. In this sense, the robust component of monetary policy is independent of the fiscal path, while the aggressive component is not—an argument for building the former, not the latter, into the reaction function.

Two dimensions of coordination lie beyond the model. First, the fiscal authority here rebates carbon revenues lump-sum; recycling revenues toward labor-tax cuts or green investment subsidies would change both the fiscal multiplier of the tax and the supply-side pressure monetary policy must absorb, plausibly reducing the value of monetary accommodation by addressing the contraction at its source. This channel is quantitatively important: Blanz et al. (2025) show, in a DSGE framework with fluctuating fossil energy prices, that the stabilization and welfare properties of market-based climate policy hinge crucially on the fiscal recycling scheme, reinforcing the conclusion that the monetary trade-offs identified here are conditional on the fiscal design. Second, the analysis takes the carbon path as exogenous to monetary policy, whereas a fiscal authority that internalizes the monetary regime might choose a different tax trajectory—for instance, phasing in the tax faster under an accommodative central bank. Modeling this strategic interaction requires a joint optimal-policy framework and is left for future research; the present results provide the monetary side of that problem, showing which components of the reaction function are robust to the fiscal environment and which are not.

6.4 The Division between Fiscal and Monetary Policy

These findings reinforce a clear division of responsibilities. The long-run trajectory of emissions is overwhelmingly determined by fiscal policy, with monetary policy having only a marginal effect on cumulative emissions. The optimal role of the central bank is therefore not to substitute for climate regulation, nor to engineer a welfare-maximizing boom that violates its mandate.

Instead, a feasible optimal policy combines a strong response to inflation with limited, state-contingent flexibility based on emissions data to smooth the transition-induced reallocation. By respecting this division, monetary policy anchors nominal expectations while allowing fiscal policy to guide the economy's structural transformation through relative price signals. In this sense, the most effective "green" monetary policy is one that preserves nominal stability and supports long-term investment during the transition.

7 Conclusions

This paper studies how monetary policy should respond to the low-carbon transition when climate policy operates through carbon taxation and financial frictions shape macroeconomic adjustment. Using a two-sector E-DSGE model with banking constraints, endogenous abatement, and recursive preferences, the analysis shows that the transition is not only a supply shock but also a financial shock, with asset revaluations and balance-sheet dynamics playing a central role in transmission.

The results highlight a sharp distinction between welfare-optimal and mandate-consistent monetary policy. Rules that aggressively accommodate declining emissions can raise welfare by improving the mean consumption path and supporting investment through the financial accelerator—even though the boom-bust dynamics they engineer generate a substantial consumption-risk penalty that claws back much of the mean-path gain. These gains arise from reallocating adjustment costs intertemporally rather than from altering the long-run emissions path, which remains pinned down by fiscal policy. However, the same mechanism produces large and persistent inflation volatility, rendering such policies incompatible with standard inflation-targeting mandates and central bank credibility.

When policy is instead evaluated through a conventional stabilization lens, the optimal response changes markedly. A small, positive reaction to emissions improves outcomes modestly without destabilizing prices or output, while preserving the standard transmission of monetary policy to demand. In this regime, emissions act as an auxiliary indicator of transition-related supply pressures rather than as an independent policy objective. Importantly, the ranking of policy rules is robust across surprise and anticipated tax implementations, across parameterizations, and to removing financial frictions, though the quantitative importance of accommodation is substantially reduced when the financial accelerator is absent.

Together, the findings support a division of roles. Fiscal policy determines the path of decarbonization through carbon pricing, while monetary policy handles how the economy absorbs the resulting effects. Central banks need not ignore the transition, but neither should they attempt to drive it directly. An approach that treats emissions dynamics as a state-contingent signal of supply stress allows monetary policy to smooth adjustment while maintaining nominal stability, ensuring that the green transition proceeds without compromising the central bank's mandate. Hence, their role is not to substitute for climate policy, but to ensure that the transition is not derailed by financial instability.

Several extensions merit separate treatment. Introducing uncertainty in the climate system—stochastic damages or ambiguous emissions paths—would interact naturally with the recursive-preference structure but requires a substantially different solution framework. Likewise, modeling strategic interaction between fiscal and monetary authorities, alternative revenue-recycling schemes, and measurement error in emissions data would refine the assessment of how much weight central banks should place on environmental indicators during the transition.

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Appendix A.1 Alternative Parametrization - Gradual Tax Shock

Table 16: Robustness of Optimal Taylor Rule Parameters - Gradual Tax

	Max Welfare			Min L_{CB}		
Parameter	ϕ_π^*	ϕ_e^*	CE (%)	ϕ_π^*	ϕ_e^*	L_{CB}
Risk Aversion (σ)						
$\sigma = 2$	1.10	0.050	-0.094	1.10	0.025	0.095
$\sigma = 10$ (baseline)	1.10	0.050	-0.094	1.10	0.025	0.095
$\sigma = 30$	1.50	0.100	-0.092	1.10	0.025	0.095
Intertemporal Elasticity of Substitution (ψ)						
$\psi = 0.5$	1.10	0.050	-0.097	1.10	0.025	0.108
$\psi = 1.5$ (baseline)	1.10	0.050	-0.094	1.10	0.025	0.095
$\psi = 2.0$	1.10	0.050	-0.094	1.10	0.025	0.094
$\psi = 5.0$	1.10	0.050	-0.094	1.10	0.025	0.091
Carbon Tax Shock Size						
0.5%	1.10	0.100	-0.040	1.10	0.025	0.090
1.0% (baseline)	1.10	0.050	-0.094	1.10	0.025	0.095
2.0%	1.10	0.050	-0.228	1.10	0.025	0.093
5.0%	1.50	0.025	-0.759	1.50	0.025	0.095
10.0%	1.50	0.000	-2.051	1.50	0.025	0.120
Habit Persistence (h)						
$h = 0.0$	1.10	0.050	-0.093	1.10	0.025	0.097
$h = 0.3$	1.10	0.050	-0.093	1.10	0.025	0.096
$h = 0.7$ (baseline)	1.10	0.050	-0.094	1.10	0.025	0.095
$h = 0.8$	1.10	0.050	-0.095	1.10	0.025	0.095
Abatement Technology						
No Abatement	1.50	0.200	-0.069	1.50	0.200	0.20
With Abatement (baseline)	1.10	0.050	-0.094	1.10	0.025	0.095
CRRA Preferences ($\psi = 1/\sigma$)						
$\sigma = 2$	1.10	0.050	-0.097	1.10	0.025	0.108
$\sigma = 10$	1.10	0.050	-0.112	1.50	0.025	0.194
Elasticity of Substitution (ξ)						
$\xi = 1.01$	1.10	0.050	-0.092	1.10	0.025	0.096
$\xi = 2.0$ (baseline)	1.10	0.050	-0.094	1.10	0.025	0.095
$\xi = 3.00$	1.10	0.050	-0.096	1.10	0.025	0.094

Notes: Optimal (ϕ_π, ϕ_e) computed over the grid $\phi_\pi \in \{1.10, 1.50\}$ and $\phi_e \in [-0.10, 0.15]$, holding $\phi_y = 0.12$ fixed. Results based on 2000 GIRF replications with order=2 pruning. CE denotes consumption-equivalent welfare. L_{CB} is the cumulative discounted central bank loss.

1206 # Appendix A.2 Transition Dynamics for Financial Variables

Figure 4: Impulse Responses to Surprise Carbon Tax Shock

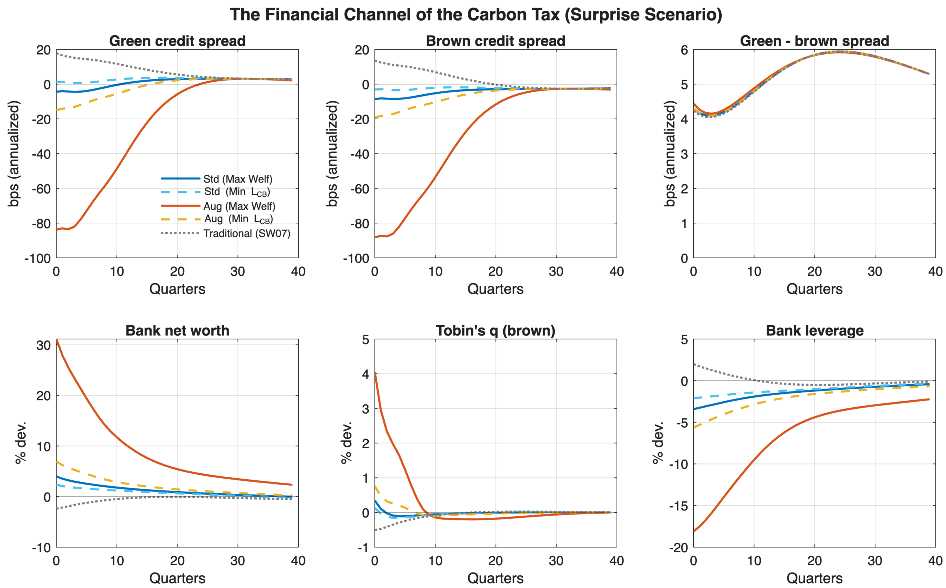
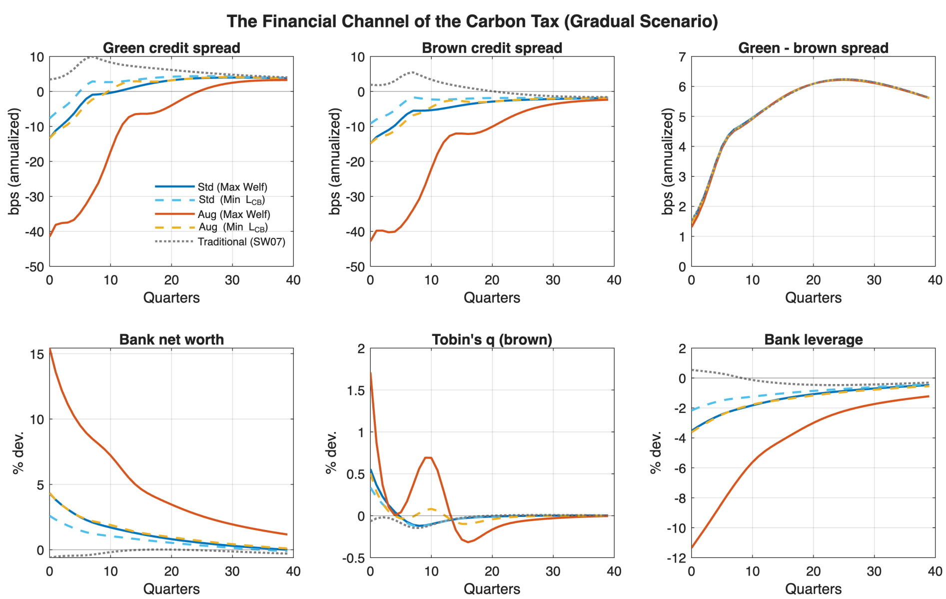


Figure 5: Impulse Responses to Gradual (Anticipated) Carbon Tax Shock



Appendix A.3 Transition Dynamics without Financial Frictions

Figure 6: Impulse Responses to Surprise Carbon Tax Shock - No Financial Frictions

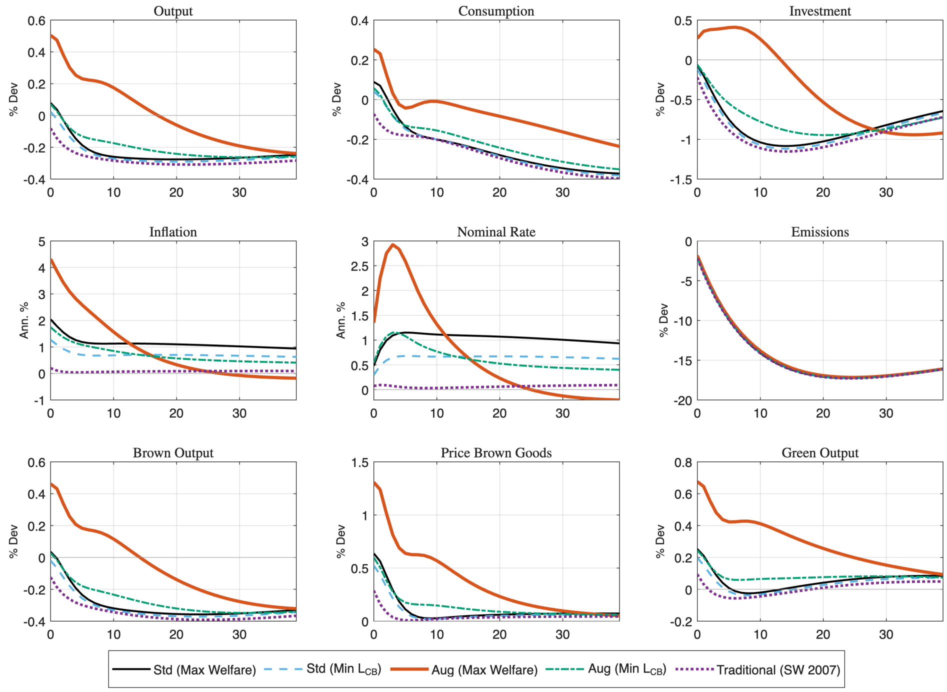
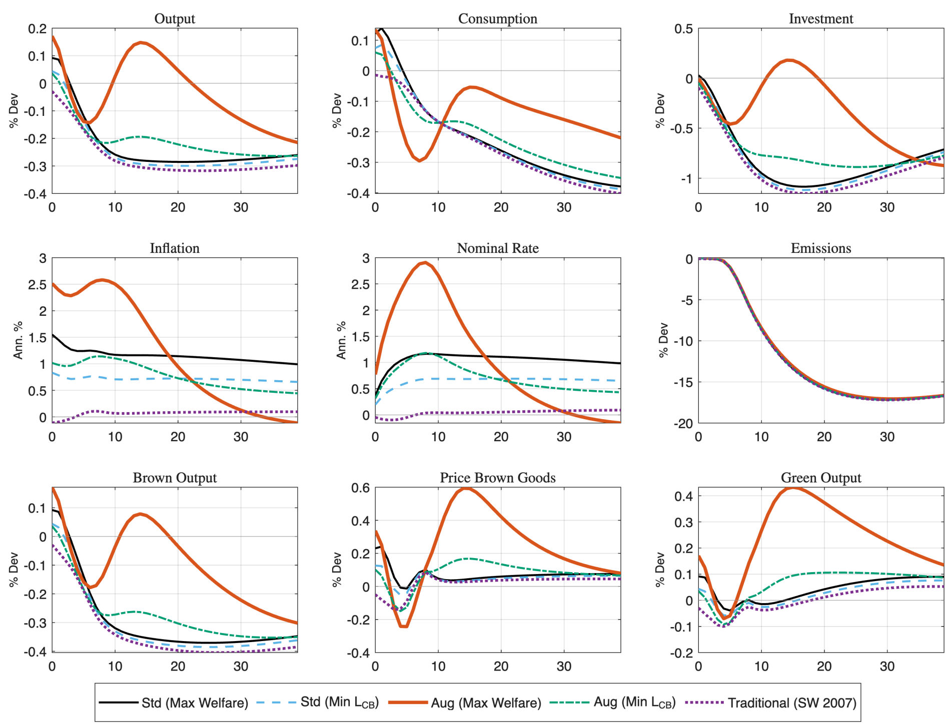


Figure 7: Impulse Responses to Gradual (Anticipated) Carbon Tax Shock - No Financial Frictions



Appendix A.4 Impulse Response Functions to a 25 bp Monetary Policy Shock

Figure 8: Impulse Responses to a Monetary Policy Shock - Real Variables

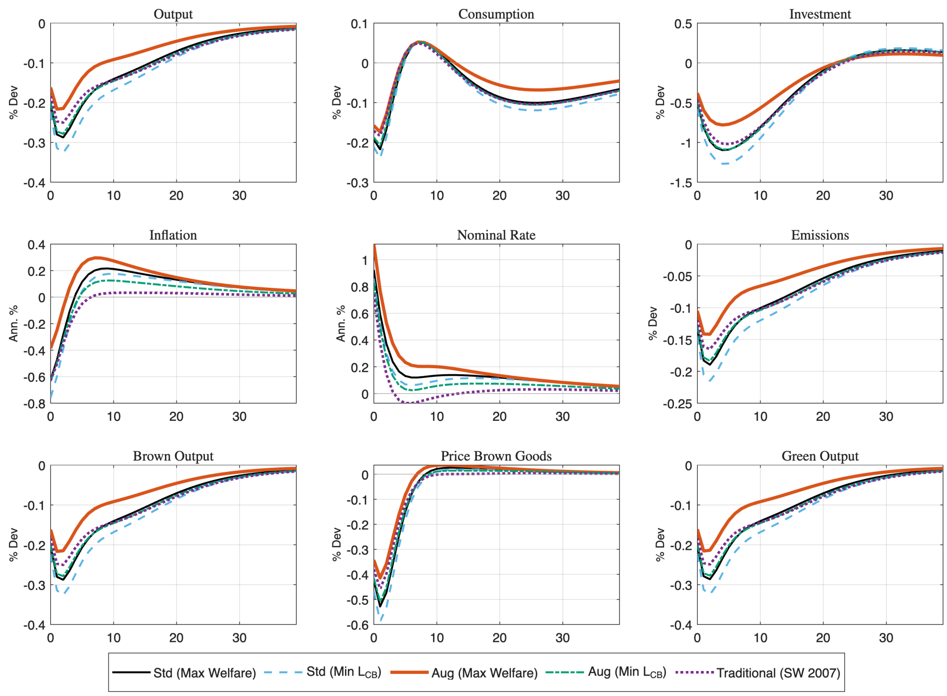
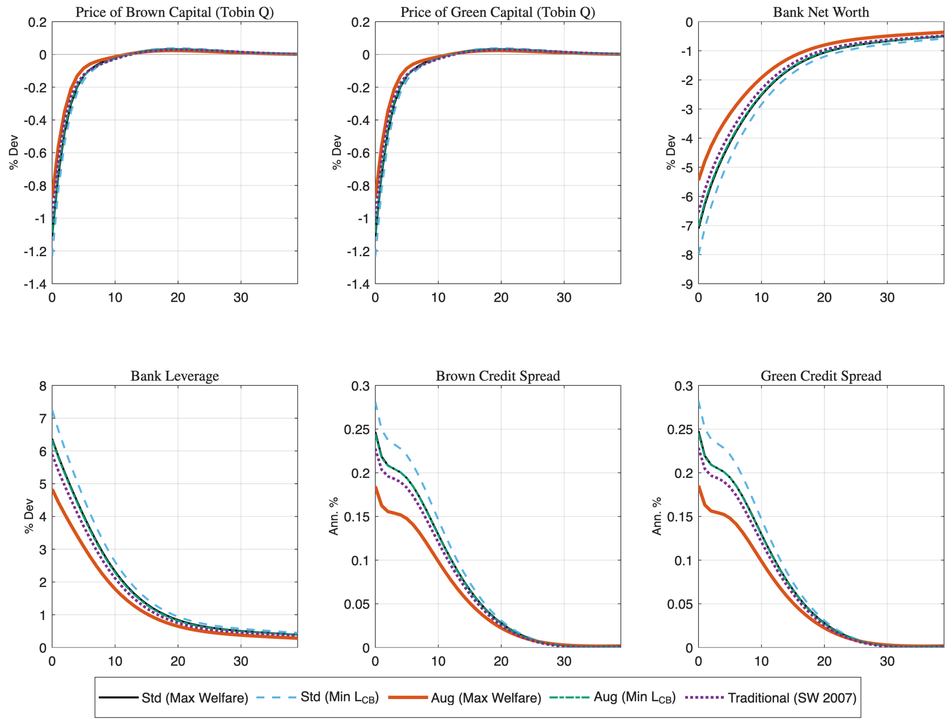


Figure 9: Impulse Responses to Monetary Policy Shock - Financial Variables



Appendix A.5 CRRA Preferences: Gradual-Shock Comparison and Impulse Responses

Table 17 reports the gradual-shock (anticipated phase-in) version of Table 15. The pattern mirrors the surprise scenario: the qualitative trade-off survives under expected utility, anticipation attenuates the peak responses of both inflation and investment, and under low-IES CRRA ($\sigma = 10$, $\psi = 0.1$) the investment channel of accommodation largely collapses (peak investment of 0.98 percent under the welfare-maximizing augmented rule, against 3.16 percent under Epstein–Zin) while inflation pressure intensifies (4.64 against 3.92 percent).

Table 17: Epstein–Zin versus CRRA Preferences: Gradual (Anticipated) Carbon Tax

Preferences	Rule	CE (%)	$L_{CB} \times 100$	e_{cum}	Peak π	Peak I
Epstein–Zin ($\sigma = 10$, $\psi = 1.5$, baseline)						
	Std (Max Welf)	−0.126	0.076	−2.883	1.05	0.20
	Std (Min L_{CB})	−0.120	0.061	−2.888	0.76	0.07
	Aug (Max Welf)	−0.109	0.135	−2.836	3.92	3.16
	Aug (Min L_{CB})	−0.114	0.042	−2.884	1.06	0.11
CRRA ($\sigma = 2$, $\psi = 0.5$)						
	Std (Max Welf)	−0.132	0.077	−2.883	1.07	0.23
	Std (Min L_{CB})	−0.124	0.060	−2.887	0.77	0.11
	Aug (Max Welf)	−0.114	0.128	−2.837	4.14	2.49
	Aug (Min L_{CB})	−0.117	0.038	−2.883	1.07	0.13
CRRA ($\sigma = 10$, $\psi = 0.1$)						
	Std (Max Welf)	−0.166	0.081	−2.880	1.16	0.30
	Std (Min L_{CB})	−0.149	0.058	−2.884	0.85	0.22
	Aug (Max Welf)	−0.135	0.115	−2.850	4.64	0.98
	Aug (Min L_{CB})	−0.130	0.031	−2.881	1.09	0.16

Notes: Anticipated phase-in (gradual) carbon tax shock; 1,000 order-2 GIRF replications, 200-quarter burn-in, as in Table 15. Peak π in annualized percentage points; Peak I in percent deviation. CRRA is nested in the recursive specification at $\sigma = 1/\psi$.

Figure 10: Impulse Responses to Surprise Carbon Tax - Real Variables - CRRA $\sigma = 10$ and $\psi = 1/10$

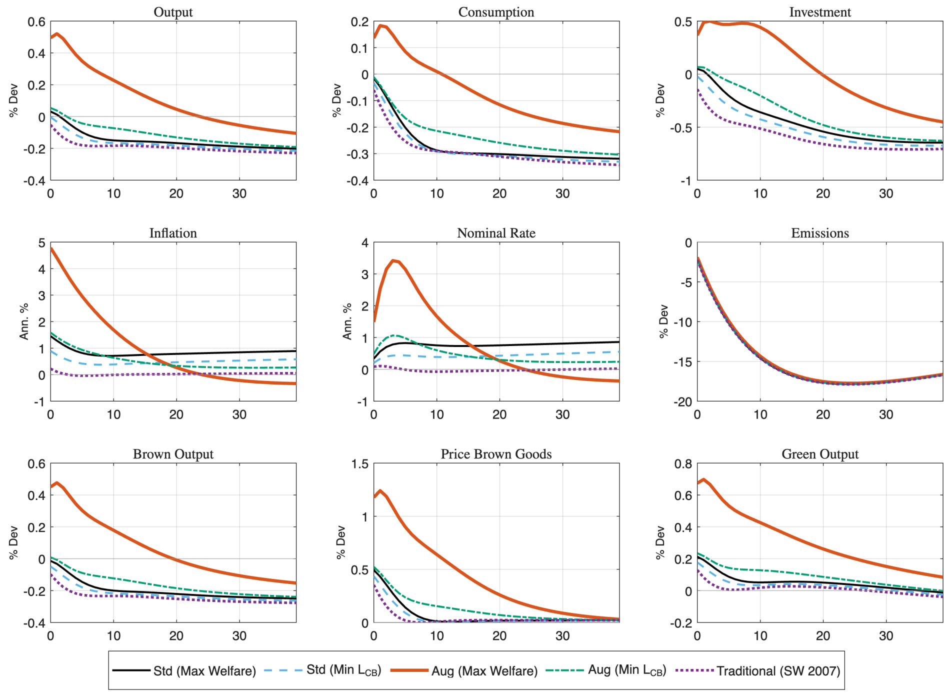


Figure 11: Impulse Responses to Surprise Carbon Tax - Financial Variables - CRRA $\sigma = 10$ and $\psi = 1/10$

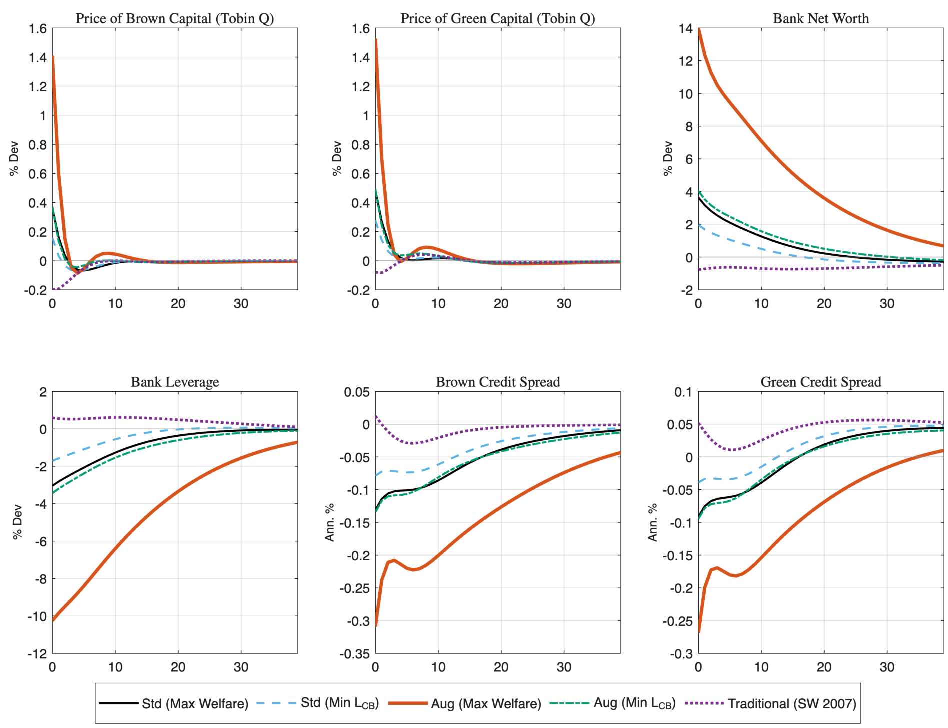


Figure 12: Impulse Responses to Surprise Carbon Tax - Real Variables - CRRA $\sigma = 2$ and $\psi = 1/2$

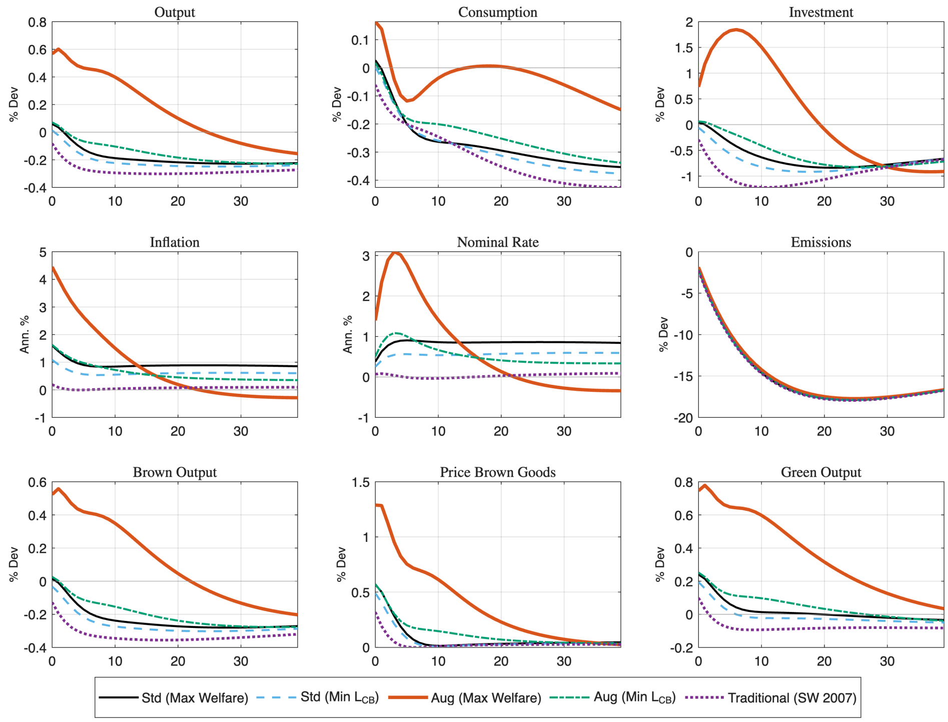
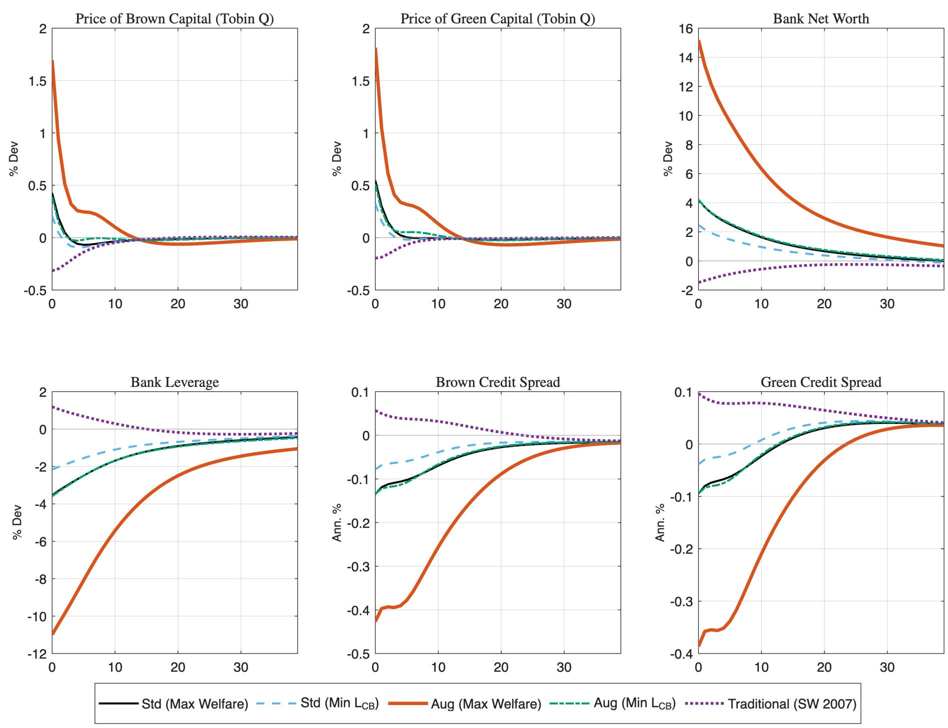


Figure 13: Impulse Responses to Surprise Carbon Tax - Financial Variables - CRRA $\sigma = 2$ and $\psi = 1/2$



Appendix A.6 Transition Dynamics under the Permanent Carbon Tax

Figure 14: Impulse Responses to a Permanent Carbon Tax - Real and Nominal Variables

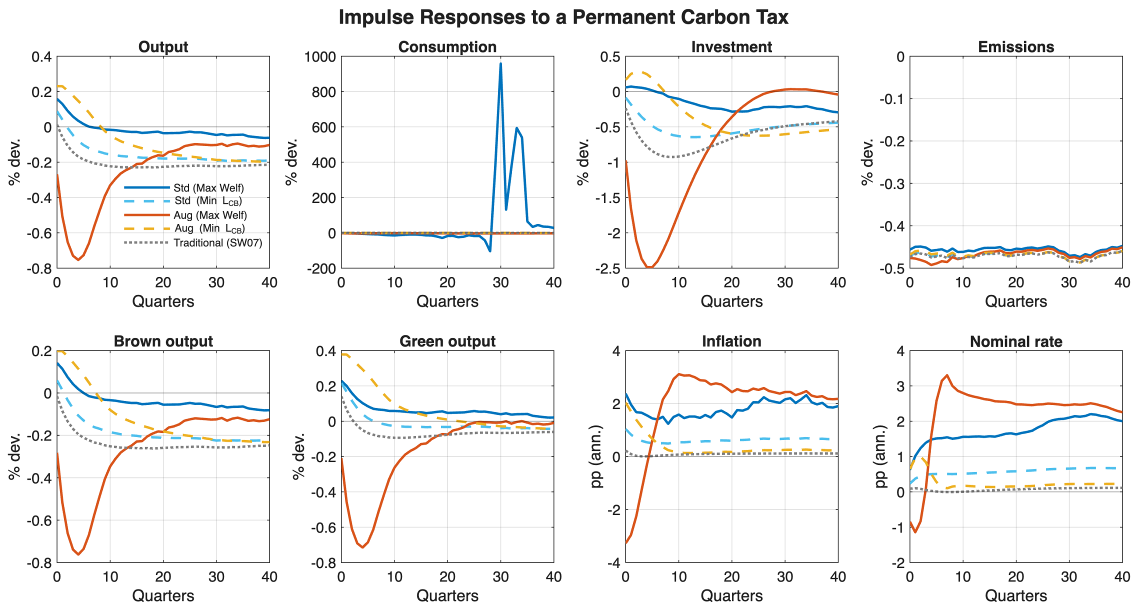
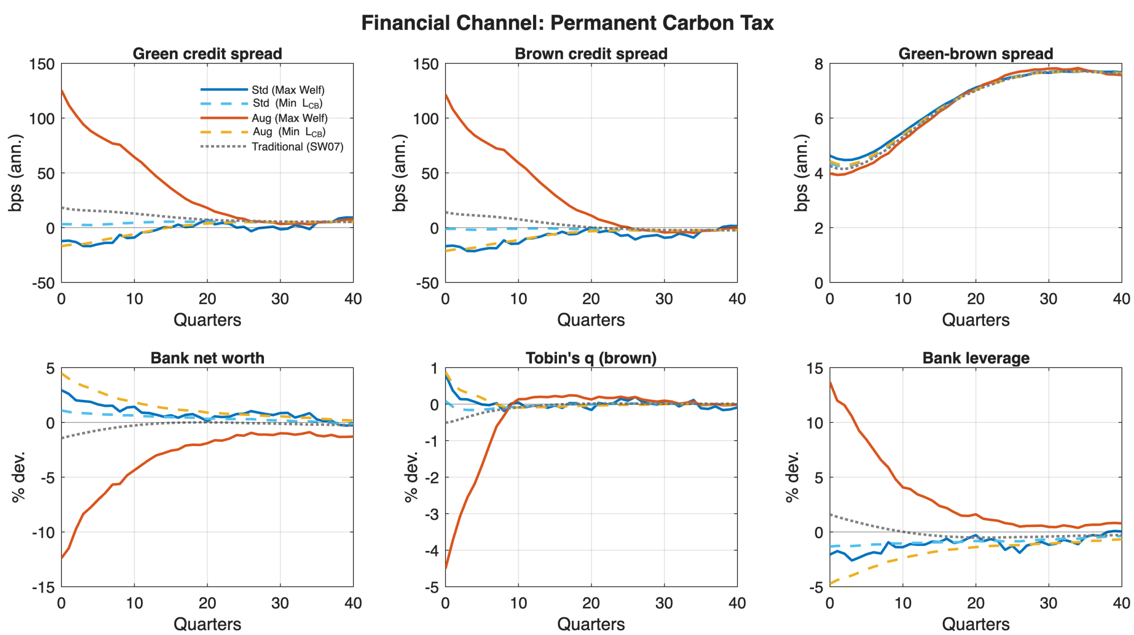


Figure 15: Impulse Responses to a Permanent Carbon Tax - Financial Variables



Appendix B.1: Calibrated Parameters

We follow Ferrari and Nispi Landi (2024) and calibrate ex-ante the parameters:

- $Lev = 6$
- $\bar{R} = \bar{\pi}/\beta = 1.005/0.995 \approx 1.010$ (a 2% annualized real rate)
- $\frac{k^G}{k} = 0.2$

And we find the ex-post parameters: $\left\{\theta, \iota, \zeta, g\right\}$

By using the net-worth evolution, we can find ι:

$$\iota = \frac{1 - \chi((r^G - R)lev + R}{lev}$$

Then, from the FOC of bankers, we can get the bank's discount factor:

$$\nu = \frac{1 - \chi}{1 - \chi\beta((r^G - R)lev + R)}$$

And from the leverage equation, we get the fraction of capital that can be diverted:

$$\theta = \frac{\nu}{lev} + \beta((r^G - R)\nu)$$

In the model, pollution is represented in arbitrary units. However, to align with real-world measurements, the aim is to calibrate the model so that the steady-state pollution level corresponds to x^{GtC} (gigatons of carbon). This choice of x^{GtC} will result in a damage parameter, denoted as $\bar{D}$, that matches the DICE-based calibration of Ferrari and Nispi Landi (2024) (following Nordhaus, 2017). By achieving this calibration, I ensure the model's pollution levels are more meaningful and relevant to actual carbon emissions and their associated impacts.

$$\overline{D} = \overline{D}_0 + \overline{D_1}x^{GtC} + \overline{D_2}\left(x^{GtC}\right)^2$$

Where $\overline{D_1}$ and $\overline{D_2}$ are taken from Ferrari and Nispi Landi (2024). We need to find D_1 and D_2 in the model. Define the emissions of the rest of the world as: $e^{row} = RoW \times \bar{e}$, and from the steady-state pollution stock ($\overline{x}$). Then we get:

$$D_1 = \overline{D_1}\frac{x^{Gtc}}{\overline{x}}, \quad D_2 = \overline{D_2}\left(\frac{x^{Gtc}}{\overline{x}}\right)$$

Appendix B.2: Steady State Equilibrium

This appendix details the system of equations characterizing the deterministic steady state of the model. In the steady state, all aggregate shocks are zero ($A_t = 1, \epsilon_t^{\tau} = 0$), and variables are constant. We denote steady-state values with a bar (e.g., $\bar{Y}$).

B.2.1 Prices and Interest Rates

The steady state is characterized by constant inflation $\bar{\pi}$. The real marginal cost $\bar{p}_I$ is determined by the monopolistic competition markup ϵ:

$$\bar{p}_I = \frac{\epsilon - 1}{\epsilon} \tag{28}$$

The nominal interest rate $\bar{R}$ is determined by the household's discount factor β and the inflation target according to the Fisher equation:

$$\bar{R} = \frac{\bar{\pi}}{\beta} \tag{29}$$

Due to financial frictions, there is a spread ($\bar{sp}$) between the risk-free rate and the return on corporate bonds. The required returns on green and brown bonds are symmetric in the steady state:

$$\bar{R}^G = \bar{R}^B = \frac{1}{\beta} + \bar{sp} \tag{30}$$

In the absence of investment adjustment costs ($\kappa_I = 0$ in steady state), the price of capital (Tobin's Q) is unity: $\bar{Q}^G = \bar{Q}^B = 1$. The rental rates of capital are thus determined by the required return and the depreciation rate δ:

$$\bar{R}^K_G = \bar{R}^G - (1 - \delta), \quad \bar{R}^K_B = \bar{R}^B - (1 - \delta) \tag{31}$$

B.2.2 Optimal Abatement and Emissions

A key feature of the model is endogenous abatement. Brown firms choose a steady-state abatement rate $\bar{\mu}$ to equate the marginal cost of abatement technology to the marginal tax savings. Given the carbon tax $\bar{\tau}$ and abatement cost parameters θ_1, θ_2:

$$\underbrace{\theta_1 \theta_2 \bar{\mu}^{\theta_2 - 1}}\text{Marginal Cost} = \underbrace{\bar{\tau}(\bar{Y}^B)^{-ps}}\text{Marginal Benefit} \tag{32}$$

Solving for $\bar{\mu}$:

$$\bar{\mu} = \left(\frac{\bar{\tau}(\bar{Y}^B)^{-ps}}{\theta_1 \theta_2} \right)^{\frac{1}{\theta_2 - 1}} \tag{33}$$

Total real abatement costs $\bar{Z}$ and net emissions $\bar{E}$ are given by:

$$\bar{Z} = \theta_1 \bar{\mu}^{\theta_2} \bar{Y}^B, \quad \bar{E} = (1 - \bar{\mu})(\bar{Y}^B)^{1-ps} \tag{34}$$

B.2.3 Production and Factor Demands

Output is aggregated from green and brown sectors using a CES technology with elasticity of substitution ξ and weight ζ. The relative prices satisfy the aggregate price index condition:

$$1 = (1 - \zeta) \left(\frac{\bar{p}_G}{\bar{p}_I} \right)^{1-\xi} + \zeta \left(\frac{\bar{p}_B}{\bar{p}_I} \right)^{1-\xi} \tag{35}$$

This implies the relative demand for sectoral output:

$$\bar{Y}^G = (1 - \zeta) \left(\frac{\bar{p}_G}{\bar{p}_I} \right)^{-\xi} \bar{Y}, \quad \bar{Y}^B = \zeta \left(\frac{\bar{p}_B}{\bar{p}_I} \right)^{-\xi} \bar{Y} \tag{36}$$

In the Green Sector, factor demands follow standard Cobb-Douglas conditions:

$$\bar{w} = (1 - \alpha) \bar{p}_G \frac{\bar{Y}^G}{\bar{L}^G}, \quad \bar{R}^K_G = \alpha \bar{p}_G \frac{\bar{Y}^G}{\bar{K}^G} \tag{37}$$

In the Brown Sector, the carbon tax and abatement costs act as a wedge on revenue. Firms pay factors from their effective price $\bar{p}_B^{eff}$, defined as the market price net of abatement costs and tax payments:

$$\bar{p}_B^{eff} = \bar{p}_B - \frac{\bar{Z}}{\bar{Y}^B} - \bar{\tau}(1-\bar{\mu})(1-ps)(\bar{Y}^B)^{-ps} \tag{38}$$

The brown factor demands are thus:

$$\bar{w} = (1-\alpha)\bar{p}_B^{eff}\frac{\bar{Y}^B}{\bar{L}^B}, \quad \bar{R}_B^K = \alpha\bar{p}_B^{eff}\frac{\bar{Y}^B}{\bar{K}^B} \tag{39}$$

B.2.4 Financial Sector

Financial intermediaries face an agency friction that limits leverage. In the steady state, the incentive constraint binds, linking the bank's value $\bar{V}^f$ to its assets. Net worth $\bar{N}$ evolves according to the accumulation of retained earnings from the spread between asset returns and deposit costs:

$$\bar{N} = \chi\left(\bar{R}^B\bar{\phi} - \bar{R}(\bar{\phi}-1)\right)\bar{N} + \iota\bar{B} \tag{40}$$

where $\bar{\phi}$ is the steady-state leverage ratio, χ is the survival probability, and ι represents transfers to new bankers. The stochastic discount factor of the banker $\bar{\nu}$ is determined by:

$$\bar{\nu} = \frac{1-\chi}{1-\chi\beta[(\bar{R}^B-\bar{R})\bar{\phi}+\bar{R}]} \tag{41}$$

The diversion parameter θ is calibrated to ensure the incentive constraint holds at the target leverage $\bar{\phi}$:

$$\theta = \frac{\bar{\nu}}{\bar{\phi}} + \beta\bar{\nu}(\bar{R}^B - \bar{R}) \tag{42}$$

B.2.5 Market Clearing

Labor markets clear such that aggregate labor $\bar{L}$ aggregates sectoral hours:

$$\bar{L} = \left[(\bar{L}^G)^{1+\rho_l} + (\bar{L}^B)^{1+\rho_l}\right]^{\frac{1}{1+\rho_l}} \tag{43}$$

The aggregate resource constraint accounts for consumption, investment, government spending, and the real resource costs of abatement:

$$\bar{Y} = \bar{C} + \bar{I} + \bar{G} + \bar{Z} \tag{44}$$

where aggregate investment covers depreciation of the total capital stock: $\bar{I} = \delta(\bar{K}^G + \bar{K}^B)$.

Appendix B.3: Banker's Problem Maximization

We solve this problem using the solution Ferrari and Nispi Landi (2024) proposed for this specific setting.

Let $M_{t,t+i}$ be the stochastic factor applying in period t to earnings at $t+i$. Each banker exits the market with probability $(1-\chi)$ getting NW_{t+1} at the beginning of period $t+i$. With probability χ, a banker continues in this activity. Therefore the value of a bank is:

$$V_t^f = \max \ E_t\left[\sum_{i=0}(1-\chi)\chi^i\beta^{i+1}M_{t,t+1}NW_{t+1+i}\right]$$

Subject to:

$$\frac{NW_t}{NW_{t-1}} = (r_t^G - r_t^B)lev_{t-1}^g + (r_t^B - \frac{r_{t-1}}{\pi_t})lev_{t-1} + \frac{r_{t-1}}{\pi_t} - \frac{\kappa_F}{2}(\frac{lev_t^g}{l_{t-1}} - b^*)^2 \tag{45}$$

Recursively writing the expected terminal wealth:

$$\frac{V_t^f}{NW_t} = \max \ E_t\left[(1-\chi) \ M_{t,t+1}\frac{NW_{t+1}}{NW_t} + \chi\beta M_{t,t+1}\frac{V_{t+1}^f}{NW_{t+1}}\right]$$

And following Gertler and Karadi (2011), financial intermediaries face the collateral constraint:

$$V_t^f(NW_t) \geq \theta b_t^f$$

$$\frac{V_t^f(NW_t)}{NW_t} \geq \theta lev_t$$

As the standard solution in this literature, we guess the solution:

$$\frac{V_t^f}{NW_t} = a_{g,t}lev_t^g + a_{b,t}lev_t + a_{c,t}(\frac{lev_t^g}{lev_t} - b^*)^2 + a_{d,t}$$

Being $a_{k,t}$ with $k = a, g, c, d$ the coefficients to be determined.

Rewriting the bank problem as:

$$\max \left[a_{g,t}lev_t^g + a_{b,t}lev_t + a_{c,t}(\frac{lev_t^g}{lev_t} - b^*)^2 + a_{d,t}\right] + (1 + \gamma_{Ft}) - \gamma_{Ft}\theta lev_t$$

With γ_{Ft} being the Lagrange multiplier. The FOC are:

$$\left[a_{g,t} + \frac{2a_{c,t}}{lev_t}(\frac{lev_t^g}{lev_t} - b^*)\right](1 + \gamma_{Ft}) = 0, \quad \left[a_{b,t} - \frac{2a_{c,t}lev_t^g}{lev_t^2}(\frac{lev_t^g}{lev_t} - b^*)\right](1 + \gamma_{Ft}) = \gamma_{Ft}\theta$$

Then, from the first FOC, we get the following:

$$a_{g,t} = -\frac{2a_{c,t}}{lev_t}(\frac{lev_t^g}{lev_t} - b^*)$$

We get the formula for γ_{Ft} from the second condition. Then, using the guessed solution, we get:

$$\frac{V_t^f}{NW_t} = \max \; E_t\left[M_{t,t+1}\nu_{t+1}\frac{NW_{t+1}}{NW_t}\right]$$

Where ν_t is the effective bank's SDF with $\nu_t = 1 - \chi + \chi\left[a_{b,t} - \frac{2a_{c,t}lev_t^g}{lev_t^2}(\frac{lev_t^g}{lev_t} - b^*)^2\right]$
Given the Banks' constraint by the evolution of net worth, we can find the remaining coefficients:

$$a_{g,t} = \mathbb{E}_t\left[M_{t,t+1}\nu_{t+1}(r_{t+1}^g - r_{t+1}^b)\right]$$

$$a_{b,t} = \mathbb{E}_t\left[M_{t,t+1}\nu_{t+1}(r_{t+1}^b - \frac{r_t}{\pi_{t+1}})\right]$$

$$a_{c,t} = \mathbb{E}_t\left[M_{t,t+1}\nu_{t+1}\frac{r_t}{\pi_{t+1}}\right]$$

$$a_{d,t} = -\frac{\kappa_F}{2}\mathbb{E}_t\left[M_{t,t+1}\nu_{t+1}\right]$$

Using the first FOC, we get the following:

$$E_t M_{t,t+1}\, \nu_{t+1}\, \left(r_{g,t+1} - r_{b,t+1} - \left(\frac{lev_{g,t}}{lev_t} - b^* \right) \frac{\kappa_F}{lev_t} \right) = 0$$

Now, we can get the respective expression for the optimal leverage:

$$\theta lev_t = a_{g,t} lev_t^g + a_{b,t} lev_t + a_{c,t}(\frac{lev_t^g}{lev_t} - b^*)^2 + a_{d,t}$$

$$lev_t = \frac{M_{t,t+1}\, \nu_{t+1}\, \left(\frac{r_t}{\pi_{t+1}} + \left(r_{g,t+1} - r_{b,t+1}\right)\, lev_{g,t} - \frac{\kappa_F}{2}\, \left(\frac{lev_{g,t}}{lev_t} - b^* \right)^2 \right)}{\theta - M_{t,t+1}\, \nu_{t+1}\, \left(r_{b,t+1} - \frac{r_t}{\pi_{t+1}} \right)}$$

Finally, using the previous conditions, we get the resulting formula for ν_t:

$$\nu_t = 1 - \chi + E_t M_{t,t+1}\, \nu_{t+1}\, \chi\, \left(\frac{r_t}{\pi_{t+1}} + \left(r_{g,t+1} - r_{b,t+1}\right)\, lev_{g,t} + lev_t\, \left(r_{b,t+1} - \frac{r_t}{\pi_{t+1}} \right) - \frac{\kappa_F}{2}\, \left(\frac{lev_{g,t}}{lev_t} - b^* \right)^2 \right)$$

Appendix B.4: Household and Capital-Producer Optimality Conditions

This appendix collects the derivations moved from the main text. The household's stochastic discount factor implied by the recursive preferences in equation (1) is

$$M_{t,t+1} = \beta \frac{c_t - hc_{t-1}}{c_{t+1} - hc_t} \frac{\left((c_{t+1} - hc_t)^{\omega}(1 - L_{t+1})^{1-\omega}\right)^{1-\rho}}{\left((c_t - hc_{t-1})^{\omega}(1 - L_t)^{1-\omega}\right)^{1-\rho}} \left(\frac{V_{t+1}^{1-\sigma}}{\mathbb{E}_t\left(V_{t+1}^{1-\sigma}\right)}\right)^{\frac{\rho-\sigma}{1-\sigma}},$$

where the last factor is the recursive-preference adjustment: it exceeds unity in states where the continuation value V_{t+1} is low relative to its certainty equivalent, raising the price of assets that pay off badly when long-run prospects deteriorate. Under expected utility ($\sigma = \rho$) the adjustment term equals one and the standard CRRA discount factor obtains.

The capital producers' first-order condition for sector $k \in \{g, b\}$ is

$$1 = q_{k,t}\left(1 - \frac{\kappa_I}{2}\left(\frac{I_{k,t}}{I_{k,t-1}} - 1\right)^2 - \kappa_I\left(\frac{I_{k,t}}{I_{k,t-1}} - 1\right)\frac{I_{k,t}}{I_{k,t-1}}\right) + \mathbb{E}_t\left[M_{t,t+1}\kappa_I q_{k,t+1}\left(\frac{I_{k,t+1}}{I_{k,t}} - 1\right)\left(\frac{I_{k,t+1}}{I_{k,t}}\right)^2\right],$$

a standard Tobin's-q relation equating the marginal cost of installing capital to its shadow price, adjusted for current and expected investment-growth adjustment costs.

Appendix C: Additional Results from the Revision

C.1 Validation: Full Moment Set Across Model Variants

Table 18 complements Table 5 in the main text by reporting relative standard deviations for the model variants used in the validation exercise.

Table 18: Model Moments Across Variants: Relative Standard Deviations

Variable	Data	Baseline	No fin. frictions
Output	1.00	1.00	1.00
Consumption	0.85	0.81	0.81
Investment	4.48	3.03	2.97
Hours	1.25	1.13	1.22
Inflation	0.58	0.43	0.44
Interest rate	0.64	0.41	0.40

Note: Standard deviations relative to output. Variants as defined in Table 5; both use the TFP and monetary shocks only. "No fin. frictions" removes credit spreads and portfolio adjustment costs.

C.2 Welfare Decomposition: Method

The decomposition in Table 9 is computed as follows. For each policy rule, the model is simulated 2,000 times with and without the carbon-tax impulse, using common draws for the remaining shocks. Let $\{c_t^{(r)}, L_t^{(r)}\}$ denote the consumption and hours paths of replication r over the 40-quarter evaluation window.

1. **Total CE** is the model-consistent consumption-equivalent change computed from the recursive value function V at impact, averaged across replications, exactly as in the main policy tables.

2. **Mean-path CE** recomputes lifetime Epstein–Zin utility by applying the model's own recursive aggregator to the *cross-replication mean* paths $\{\bar{c}_t, \bar{L}_t\}$ (with the continuation value at the terminal date approximated by the stationary value of the terminal flow; the truncation is common to all rules and differences out). Averaging before evaluating eliminates the dispersion across stochastic paths, so this object isolates the welfare effect of the average trajectory.

3. **Leisure component**: the mean-path CE is recomputed holding hours at the no-shock mean

path; the difference between the full mean-path CE and this object attributes the contribution of the hours response.

4. **Consumption-risk component**: total CE minus mean-path CE. This is the part of the welfare effect coming from the dispersion of paths across states of the world, which the recursive preferences price.

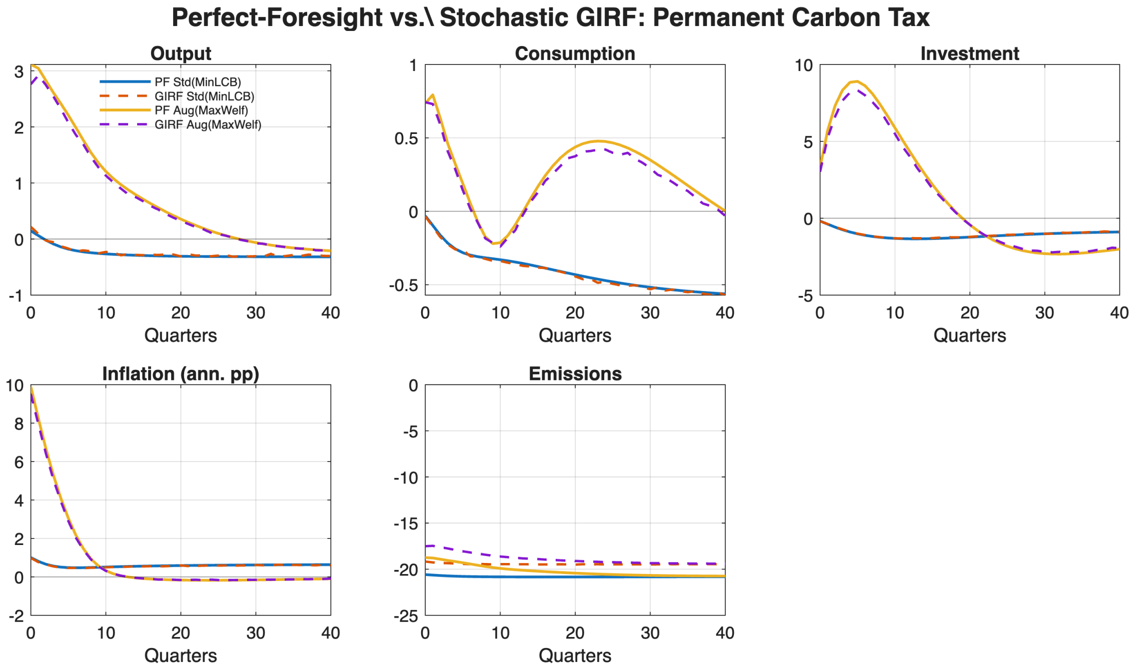
5. **Environmental component**: the emissions-path difference is cumulated into the pollution stock, converted into TFP units through the damage function, and annuitized into consumption-equivalent units.

Components sum to the total up to a small approximation residual (below 0.001 percentage points for all rules), reported in the replication code.

C.3 Perfect-Foresight Check of the Permanent-Tax Specification

Figure 16 compares the stochastic generalized impulse responses under the quasi-permanent carbon tax ($\rho_\tau = 0.9999$) with a deterministic perfect-foresight simulation of the same single-impulse experiment over a long horizon (800 quarters; by the turnpike property, the first 40 quarters are unaffected by the remote terminal condition). The two solution methods deliver essentially identical real dynamics over the evaluation window, confirming that the quasi-permanent stochastic specification is an accurate implementation of a permanent policy change.

Figure 16: Perfect-Foresight vs. Stochastic GIRF: Permanent Carbon Tax



Solid lines: deterministic perfect-foresight paths. Dashed lines: mean generalized impulse responses from the stochastic order-2 solution. Rules: loss-minimizing standard rule and welfare-maximizing augmented rule under the permanent tax.